VAJIRAM & RAVI

Institute for IAS Examination

PowerUp

Current Affairs Test Series – 2025

CA Test – 10 – (CA2260)

Answer Key &
Detailed Answer Explanations

CA Test 10 - Answer Key									
1. (c)	11. (c)	21. (d)	31. (b)	41. (c)	51. (d)	61. (b)	71. (a)	81. (a)	91. (c)
2. (b)	12. (b)	22. (d)	32. (b)	42. (c)	52. (a)	62. (c)	72. (d)	82. (b)	92. (d)
3. (d)	13. (d)	23. (b)	33. (b)	43. (a)	53. (d)	63. (d)	73. (b)	83. (a)	93. (b)
4. (b)	14. (c)	24. (d)	34. (c)	44. (b)	54. (d)	64. (c)	74. (c)	84. (d)	94. (a)
5. (b)	15. (a)	25. (c)	35. (b)	45. (c)	55. (c)	65. (c)	75. (d)	85. (d)	95. (a)
6. (b)	16. (b)	26. (d)	36. (c)	46. (c)	56. (a)	66. (b)	76. (a)	86. (c)	96. (d)
7. (a)	17. (c)	27. (c)	37. (b)	47. (b)	57. (c)	67. (a)	77. (c)	87. (d)	97. (d)
8. (c)	18. (b)	28. (c)	38. (b)	48. (b)	58. (b)	68. (a)	78. (b)	88. (c)	98. (b)
9. (d)	19. (c)	29. (c)	39. (b)	49. (a)	59. (c)	69. (b)	79. (d)	89. (a)	99. (a)
10. (d)	20. (c)	30. (c)	40. (c)	50. (b)	60. (c)	70. (b)	80. (a)	90. (b)	100. (c)

Q1.
Answer: c
Explanation:

- **Stakeholder capitalism** is a system of corporate governance, which is **oriented to serve the interests of all their stakeholders (customers, suppliers, employees, shareholder local communities).**
- It is in contrast to the prevalent principles of **shareholder capitalism**, which emphasize that executives only work for the owners and the only social responsibility of the firm is to stay within the rules of the game.
- The Stakeholder model **promotes corporate social responsibility, sustainability, and ethical governance.** It is increasingly seen as a way to balance economic success with social progress, helping address global issues like climate change, labor rights, and economic disparity.
- Other Options:
    - **Shareholder Capitalism**: It focuses on maximizing financial returns for investors and shareholders. In this system, companies prioritize profits above all else, often emphasizing short-term financial gains over long-term sustainability or social responsibility.
    - **State Capitalism:** An economic model where the government plays a central role in economic planning and decision-making. In this system, industries may be heavily regulated, or key sectors of the economy might be state-owned.
    - **Entrepreneurial Capitalism**: This form of capitalism is driven by private individuals who take risks to innovate and build businesses with minimal government interference. This system, prevalent in economies like the United States, encourages technological advancements, job creation, and competition. It relies on free-market principles where entrepreneurs and startups play a crucial role in economic development.

Therefore, option (c) is the correct answer.

Q2.
Answer: b
Explanation:

- **Olive Ridleys** are the second smallest and the most abundant sea turtle species. However, their nesting sites are concentrated in limited locations, making them highly vulnerable to habitat loss, coastal development, and environmental changes.
- The **Gahirmatha Beach (Odisha, India) is one of the world's largest nesting sites**, where hundreds of thousands of turtles gather for *arribada* (mass nesting).
- These turtles are listed as **Vulnerable** under the IUCN Red List. Their populations are declining due to factors such as incidental capture in fishing gear, loss of nesting sites, and climate change, but they are still the most abundant among all sea turtle species. **So, statement 3 is not correct.**
- Like all sea turtles, **they are air-breathing reptiles that must frequently come to the surface for breathing.** During this exercise the turtles become vulnerable to be trapped in trawl nets or gill nets. This is a major threat to their survival, leading to the call for turtle-excluder devices (TEDs) in fishing nets. **So, statement 1 is correct.**
- **Sea turtles exhibit temperature-dependent sex determination (TSD),** where the temperature of the nesting site determines the sex of hatchlings. **Warmer temperatures lead to a higher proportion of female hatchlings, while cooler temperatures produce more males.** Rising global temperatures are skewing this balance, leading to fewer males, this could impact future breeding success and long-term population stability. **So, statement 2 is correct.**

So, only two of the statements given above are correct.
Therefore, option (b) is the correct answer.
Relevance: A large number of Olive Ridley Turtle carcasses were recently found along the Tamil Nadu coast, raising serious conservation concerns.

Q3.
Answer: d
Explanation:

- **Article 174 of the Constitution of India** gives the Governor the power to summon from time to time "the House or each House of the Legislature of the State to meet at such time and place as he thinks fit…"
- However, the phrase "as he thinks fit" is read along with Article 163 of the Constitution which says that the **Governor acts on the aid and advice of the Council of Ministers**. Article 163(1) essentially limits any discretionary power of the Governor only to cases where the Constitution expressly specifies that the Governor must act on his own and apply an independent mind.
- In 2016, a Constitution Bench of the Supreme Court in ***Nabam Rebia and Bamang Felix vs Deputy Speaker*,** expressly said that the **power to summon the House is not solely vested in the Governor**. **So, statement 1 is not correct.**
- There are varying terms used for the suspension of a sitting legislature.
    - **Adjournment:** This term is used for suspending a sitting of legislature, with a predefined time and place for reassembly.
    - **Adjournment *sine die*:** This term is used when any sitting of legislature is suspended, without declaring the time and place for the next sitting or assembly. This term is usually used for the adjournment of the last sitting of the session.
    - **Prorogation:** This term denotes the termination of a session of the legislature. Though the adjournments are called by the Speaker of the assembly, prorogation is called by the President or the Governor post the Adjournment *sine die* is called by the Speaker.
- As declared by the Supreme court in ***State of Punjab Vs. Principal Secretary to the Governor of Punjab and Another***, the Speaker holds the authority to reconvene the session if the House has been

adjourned *sine die* without formal prorogation by the Governor. This was reiterated recently by the Delhi High Court, stating that the timing of reconvening a sitting after adjournment *sine die* is the **sole prerogative of the Speaker, not the Governor**. **So, statement 2 is correct.**

So, Statement-I is not correct but Statement-II is correct.
Therefore, option (d) is the correct answer.
Relevance: Delhi HC recently rejected the appeal of Delhi legislators to convene a special session of Delhi Vidhan Sabha to lay the CAG reports in assembly.

Q4.
Answer: b
Explanation:

- **Sunspots** are dark, planet sized regions that appear on the surface of the sun. They are the result of twisting, chaotic magnetic fields from within the sun's convective zone. These powerful magnetic disturbances produce active regions on the sun, which often create solar flares and CMEs.
- Sunspots are regions of **intense magnetic activity**. **These strong magnetic fields inhibit convective heat transfer**, preventing hotter plasma from rising to the surface. **So, statement 2 is not correct.**
- **The magnetic suppression results in sunspots being cooler than the surrounding areas.** The temperature of a typical sunspot is around 3,500°C (6,500°F), while the surrounding solar surface, or photosphere, is about 5,500°C (10,000°F). Due to this temperature difference, **sunspots appear darker** against the brighter background of the Sun. **So, statement 1 is correct.**
- Sunspots are temporary phenomena, evolving over days to months before dissipating. Their number and intensity fluctuate with the 11-year solar cycle, with **sunspots being more prevalent during the solar maximum and fewer or absent during the solar minimum.** Over time, they shift in location, generally forming at mid-latitudes at the start of a cycle and moving toward the equator as the cycle progresses. **It is often inferred that the number of sunspots is positively correlated with the activity of the sun (solar flares, solar winds, CMEs etc.)**. **So, statement 3 is correct.**

So, only two of the above given statements are correct.
Therefore, option (b) is the correct answer.
Relevance: During ISRO's Space Docking experiment, scientists were closely monitoring Coronal Mass Ejections (CMEs), which are closely linked to the increase in sunspots.

Q5.
Answer: b
Explanation:

- **Project Nexus**, a multilateral international initiative to enable instant **cross-border retail payments by interlinking domestic Fast Payments Systems (FPSs).**
- It is conceptualized by the Innovation Hub of the Bank for International Settlements (BIS). It seeks to enhance cross-border payments by connecting multiple domestic instant payment systems (IPS) globally. It is the first BIS Innovation Hub project in the payments area to move towards live implementation.
- Other Options:
    - **Project Mariana**: It is a mechanism by BIS to facilitate cross-border trading and settlement of Central Bank Digital Currencies (CBDCs) between financial institutions, using new Decentralized Finance (DeFi) technology.

- **Operation Twist:** A tool used by the Central Banks to manipulate bond yields and inflation by selling long term maturity bonds and purchasing short term bonds in lieu from the markets.

Therefore, option (b) is the correct answer.

Q6.
Answer: b
Explanation:

- **Ratnagiri is one of Odisha's most important Buddhist heritage sites in India,** dating back to the 5th-13th century. It flourished particularly **under the Bhaumakara dynasty (8th-10th century),** which promoted Buddhism in Odisha. The site contains numerous votive stupas, sculptures, and monasteries, reflecting its deep religious and historical significance.
- Ratnagiri was a prominent centre for ***Mahayana*** and ***Vajrayana* Buddhism** [not *Hinayana*]. The site contains numerous sculptures and inscriptions depicting Bodhisattvas, tantric deities, and Vajrayana rituals. Excavations have revealed several monasteries and stupas associated with esoteric Buddhist practices. This confirms that Ratnagiri played a crucial role in the evolution of later Buddhist traditions. **So, statement 1 is correct.**
- Ratnagiri is part of the **Diamond Triangle of Odisha**, which includes the Buddhist sites Udaygiri and Lalitgiri as well. **[*The Golden Triangle refers to Bhubaneswar, Puri, and Konark, which are major Hindu temple sites in Odisha*].** The Diamond Triangle is crucial in understanding the spread of Buddhism in eastern India. These sites together provide insights into the transition from early Buddhist traditions to later Vajrayana influences. **So, statement 2 is not correct.**
- The monastery in Ratnagiri is distinct due to its **curvilinear roof**, which is an unusual feature in Buddhist monastic architecture. **Most Buddhist monasteries have flat or gently sloping roofs, making this an architectural rarity.** This unique design suggests influences from local and possibly tantric architectural styles. The structure highlights Ratnagiri's importance as a major Buddhist educational and religious hub. **So, statement 3 is correct.**
- Recent Archaeological Survey of India's excavations at Ratnagiri have uncovered a colossal Buddha head and a giant palm, suggesting large-scale Buddhist artistry.

So, only two of the above given statements are correct.
Therefore, option (b) is the correct answer.
Relevance: ASI have renewed their excavation efforts at Ratnagiri, Odisha. This site is believed to be the rival of Nalanda as a Buddhist learning site.

Q7.
Answer: a
Explanation:

- The **Israel-Palestine conflict** is one of the longest running conflicts of the modern day history of the world. Having its genesis in the 1948 international resolution for the creation of Israel as a nation for Jewish people, which was vehemently opposed by the Arab brethren and the Islamic countries.
- Key events of this conflict:
  - **Six-Day War (1967):** It was fought between Israel and the Arab states led by Egypt, Jordan, and Syria. The war, which continued for six days, oversaw Israel gaining several advances, it occupied the territories of Sinai Peninsula, Golan heights, Gaza strip and the West bank. This war significantly altered the geopolitical landscape of the Middle East, leading to territorial disputes which are still ongoing. Following the war, the United Nations passed Resolution 242, calling for Israeli withdrawal from occupied territories in exchange for peace.
  - **Camp David Accords (1978):** These were peace brokering agreements mediated by the US. Under these accords, Israel agreed to return the Sinai Peninsula to Egypt, in return the latter

agreed to recognize Israel as a sovereign nation and cease all hostilities towards it, thus Egypt became the first Arab nation to give recognition to Israel.

- **Oslo Accords (1993):** These established direct negotiations between Israel and the Palestine Liberation Organization (PLO). The agreement facilitated the creation of the Palestinian Authority (PA) with limited self-rule in parts of the West Bank and Gaza Strip. This was the first instance of mutual recognition between Israel and the PLO.
- **Abraham Accords (2020):** These represented a new diplomatic shift in the Middle East, with Israel normalizing relations with the United Arab Emirates, Bahrain, Sudan, and Morocco under the mediation of the United States. Unlike previous peace agreements, these accords did not require an Israeli withdrawal from occupied Palestinian territories or a resolution to the Israeli-Palestinian conflict. Instead, they focused on economic, security, and technological cooperation.

So, the correct chronological sequence for the given events is Six-Day War - Camp David Accords - Oslo Accords - Abraham Accords.
Therefore, option (a) is the correct answer.
Relevance: Israel and the Hamas group have agreed for a ceasefire and return of hostages, bringing end to a two-year active conflict.

Q8.
Answer: c
Explanation:

- **An atomic clock** is a device that uses **an internal resonance frequency of atoms or molecules to measure the passage of time**. Atomic clocks are relied on internationally as the most accurate time and frequency standards with myriad uses in commerce, industry, and science.
- Most modern clocks keep time using a quartz crystal oscillator. These devices take advantage of the fact that quartz crystals vibrate at a precise frequency when voltage is applied to them. The vibrations of the crystal act like a pendulum, ticking off how much time has passed. **Atomic clocks combine a quartz crystal oscillator with an ensemble of atoms to achieve greater stability.**
- Atoms are composed of a nucleus (consisting of protons and neutrons) surrounded by electrons. Each element on the periodic table represents an atom with a certain number of protons in its nucleus. The number of electrons swarming around the nucleus can vary, but they must occupy discrete energy levels, or orbits.
- When provided energy, in atomic clocks through microwaves, can cause an electron to rise to a higher orbit around the nucleus. **The electron must receive exactly the right amount of energy — meaning the microwaves must have a very specific frequency — in order to make this jump.**
- The energy required to make electrons change orbits is **unique in each element and consistent throughout the universe for all atoms of a given element.**
- The fact that **the energy difference between these orbits is such a precise and stable value is the key ingredient for atomic clocks**; this regular oscillation is then taken as a reference to measure time.
- Although, just like any other clock, **the atomic clocks too loose or gain time**, the value is highly negligible, to the tune of a microsecond (one millionth of a second) after 10 years. This is equivalent to being off by only one second every 10 million years.

Therefore, option (c) is the correct answer.
Relevance: ISRO recently deployed an atomic clock on the NVS-02 satellite, a first for ISRO.

Q9.
Answer: d
Explanation:

- **Article 224A of the Constitution of India** provides the legal framework for appointing retired judges to act as Judges in High Courts. These provisions are used in case of the lack of quorum for the bench or increased backlog of cases.
- As per the Article, **the Chief Justice of a High Court, with the prior consent of the President,** can request a retired judge of that or any other High Court to sit as a judge. **The Governor does not play a role in this process**, as it is exclusively between the Chief Justice and the President. **So, statement 1 is not correct.**
- There is **no fixed constitutional limit on the tenure of a retired judge appointed to act as a High Court judge**. While the Supreme Court has suggested terms of 2-3 years in certain cases, the duration is not constitutionally restricted. The appointments are typically made based on the needs of the High Court and the availability of retired judges. **So, statement 2 is not correct.**
- As noted and reiterated by the Supreme Court, the provision in Article 224A allowing the appointment of retired judges has rarely been utilized. There have only been a few recorded instances where retired judges have been appointed on an ad hoc basis, making it a **dormant provision that has not been actively used in practice,** despite its potential to help reduce case backlogs.

Therefore, option (d) is the correct answer.

Knowledge Box
Difference between Retired, Acting and Ad-hoc judges: Although all these three terms are used interchangeably, but under the constitutional terms these three have varied meanings. ***Ad-hoc* Judges**: Article 127 belonging to Chapter IV of the Part V of the Indian constitution dealing with the Supreme Court, mentions the term of Ad-hoc Judge. Hence, any judge appointed to the bench of the Supreme Court for a temporary basis can be called as an *Ad-hoc* judge. **Acting Judge**: Article 126 (Part V) and Article 223 (Part VI) of the Indian constitution provides for Acting Chief Justices of the Supreme Court and High Court respectively. Further, Article 223 provides for the appointment of Acting judges and Additional judges for the High Court. **Retired Judges:** Article 128 and Article 224A provides for retired judges to be appointed on the bench of Supreme Court and High Court respectively. **Aspirants must note here that though the term Ad-hoc is being used by the Apex court in its latest statement regarding the appointment of Retired judges for High Courts, but it is a constitutional misnomer.*

Relevance: The Supreme Court recently proposed appointing retired High Court judges on an ad hoc basis to tackle the growing backlog of cases.

Q10.
Answer: d
Explanation:

- A cell is the smallest structural and functional unit of the body. Every cell contains the full set of that individual's genetic material which comprises numerous genes. Characters or traits are inherited from parents to offspring through these genes, which contain information for the production of proteins. **A cell's function and behavior depend on which genes are activated**, and so which proteins are produced.
- Different genes are activated in different cells, depending on that cell's role in the body. As a cell becomes more specialized within a tissue type, that is it assumes its specific role, there is no more need for the cell to generate proteins which are not serving its function despite having the genes for the same. **Hence as the specialization of the cell grows the number of genes it can activate decreases**. **So, option (a) is correct.**
- The DNA (or RNA) as genetic material is packed inside the cell by means of proteins known as histones. Such a packed structure is known as chromatin, which acts as a determinant of cellular activity.
- Two types of chromatin are usually found inside a cell – **tightly packed heterochromatin** and **loosely packed Euchromatin**. For the active expression of genes, it needs to gain access to specific biomolecules, these biomolecules can readily attach to part of chromatin that is loosely bound. Hence, **the metabolically active cells are predominated by Euchromatin**. **So, option (b) is correct.**
- The **chromatin is further condensed to form chromosomes at the time of cell division**. These chromosomes are usually paired in higher organisms (diploid organisms) with each set of chromosomes derived from one parent. **The cells involved in sexual reproduction (gametes – sperms and ova) contain only one set of chromosomes which gets passed to the offspring**. **So, option (c) is correct.**
- The majority of the DNA (over 98%) is found in the nucleus of the cell. But, DNA is also found in mitochondria. A small amount of DNA is found in the **mitochondria (chloroplast in plants)**, where it is called mitochondrial DNA or mtDNA. **So, option (d) is not correct.**

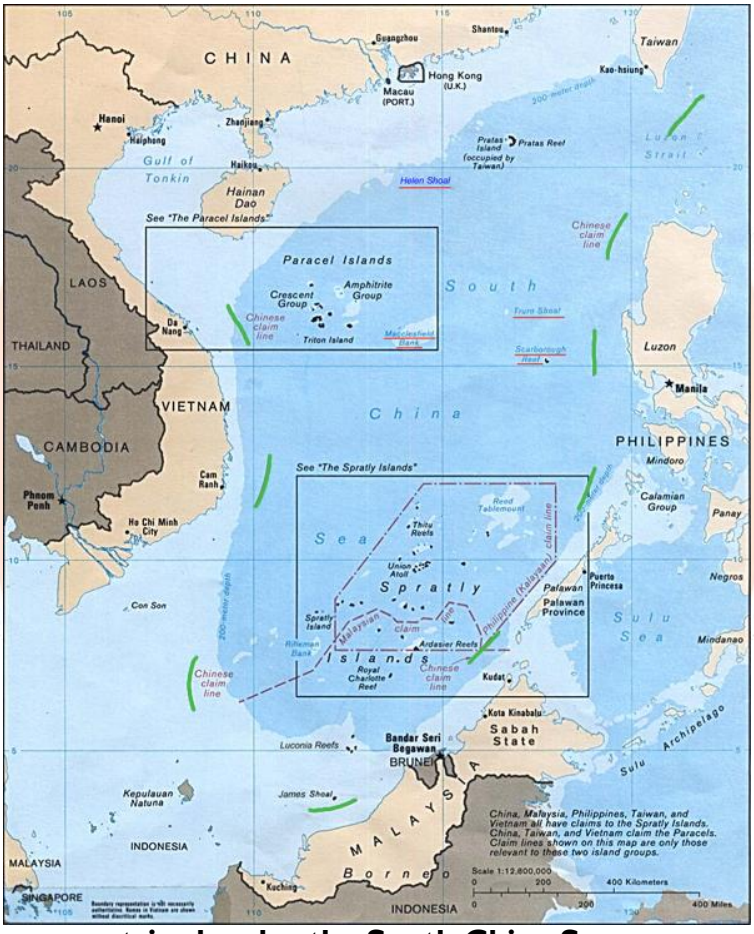
Therefore, option (d) is the correct answer.
Relevance: Prime Minister's Science, Technology & Innovation Advisory Council (PM-STIAC) Meeting discussed Cell and Gene Therapy in India

Q11.
Answer: c
Explanation:

- The **South China Sea** is a marginal sea in the western Pacific Ocean. It encompasses roughly from Singapore and the Strait of Malacca in the southwest, to the Strait of Taiwan (between Taiwan and mainland China) in the northeast.
- The area includes hundreds of small islands, rocks and reefs, the majority of which are located within the Paracel and Spratly island chains. Most of these islands are not arable, do not support permanent crops and have no meadows, pastures or forests. But the surrounding **water areas are abundant in oil, natural gas, minerals and seafood.**
- The South China Sea is bordered by multiple countries in Southeast Asia. These include:

- **Indonesia** (specifically the Natuna Islands). **So, point 1 is correct.**
- Brunei
- **Philippines** (archipelago nation with extensive maritime claims). **So, point 2 is correct.**
- Malaysia
- **Taiwan** (an island claimed by China, situated in the northern part of the South China Sea). **So, point 4 is correct.**
- **Vietnam** (with an extensive coastline facing the sea). **So, point 5 is correct.**
- Singapore

- Laos PDR, is a landlocked country and does not have a coastline, making it the incorrect option in this list. **So, point 3 is not correct.**



So, only four of the above countries border the South China Sea.
Therefore, option (c) is the correct answer.

Knowledge Box
South China Sea Dispute: **The Nine-Dash Line** China's Nine-Dash Line is an expansive claim over nearly 90% of the South China Sea, based on historical maps dating back to 1947. Originally, it was the Eleven-Dash Line, but it was reduced to nine dashes after China ceded some areas to Vietnam in 1953. **Disputed Islands and Territorial Claims in South China Sea:** • **Paracel Islands**: Disputed between China and Vietnam. China occupies them.

- **Spratly Islands:** Disputed among China, the Philippines, Vietnam, Malaysia, Brunei, and Taiwan.
- **Scarborough Shoal**: Contested between China and the Philippines.
- **Woody Island**: Part of the Paracel chain, controlled by China but claimed by Vietnam and Taiwan.
- **Natuna Islands**: Controlled by Indonesia, but near China's claimed waters, leading to occasional disputes.

Strategic Importance of the South China Sea:

- Holds 11 billion barrels of oil and 190 trillion cubic feet of natural gas.
- Crucial trade route, 21% of global trade transits through these waters.
- Rich fishing grounds support millions of livelihoods in the region.

International Rulings and Current Status:

- In 2016, an international tribunal (UNCLOS ruling) rejected China's Nine-Dash Line, favoring the Philippines.
- Despite the ruling, China has continued militarizing artificial islands and extending its naval presence.
- The United States and regional nations (India, Japan, ASEAN) have urged China to follow international laws and avoid unilateral actions.

This dispute remains a key geopolitical flashpoint, involving China, ASEAN nations, and global powers like the US and India.

Relevance: Indonesian President in his recent visit to India as the chief guest for Republic Day, in a joint statement with his Indian counterpart underscored the need for Code of Conduct in South China Sea.

Q12.
Answer: b
Explanation:

- **Nuclear Fusion is the process in which two light atomic nuclei combine to form a heavier nucleus,** releasing immense amounts of energy. This is the same process that powers the Sun and other stars. It is distinct from Nuclear Fission, where a large atomic nucleus splits into smaller ones.
- Fusion does not involve splitting atoms but rather the **merging of lighter nuclei (such as hydrogen isotopes) to form heavier ones (such as helium)**. Fission, not fusion, is the process that involves splitting of atoms. **So, statement 1 is not correct.**
- **Fusion releases far greater energy than fission**. This is due to the **binding energy per nucleon principle**, where fusing smaller nuclei results in more energy being released compared to breaking apart heavy nuclei. **So, statement 2 is correct.**
- Fusion reactions require very high temperatures, hundreds of millions of degrees Celsius — higher than the temperatures in the Sun's core. At such high temperatures, **matter exists only in the plasma state**, in which atoms get split into positively and negatively charged particles. Such hot plasma cannot be handled by or contained in any material. Within the reactor, this plasma needs to be

kept suspended in a confined space, surrounded by very strong magnetic fields acting as walls. Charged particles respond to magnetic fields, and this property is used to guide the flow of plasma within an enclosed space, separated from any matter.

- To sustain the nuclear fusion reaction, a magnetic confinement device is required. The most common types of magnetic confinement devices are **tokamaks and stellarators. These** devices maintain extreme temperatures and pressure conditions required for fusion. **So, statement 3 is correct.**

So, only two of the above statements are correct.
Therefore, option (b) is the correct answer.

Knowledge Box:
International Thermonuclear Experimental Reactor: ● ITER (International Thermonuclear Experimental Reactor) is a global collaboration designed to demonstrate the feasibility of sustained nuclear fusion as a clean energy source. ● It is the world's largest experimental fusion reactor, under construction in France, along with participation from China, India, the EU, the US, Russia, South Korea, and Japan. ● If successful, ITER will pave the way for commercial fusion reactors in the future, potentially solving the global energy crisis.

Relevance: China has recently announced the successful conduct of a Tokamak experiment, paving the way for a controlled Nuclear Fusion reaction in the future.

Q13.
Answer: d
Explanation:

- **The Adi is a major tribal group**. They live within the districts of Siang, East Siang, Upper Siang, parts of West Siang, Lower Dibang Valley, Lohit, Shi Yomi and Namsai in **Arunachal Pradesh.**
- Adi is a collective term for groups Padams*, Milangs, Komkars, Minyongs* and *Pasis* who collectively call themselves *Adi*, meaning hill people. Akin to one another, they speak the same dialect, claim a common origin and also perform and celebrate the same rituals and festivals.
- The *Adis* are agriculturalists. Both wet rice cultivation and shifting cultivation are practiced. Dry-lands are used to grow cash crops like Maize and Mustards. They are revered for their traditional dance ***Ponung***.

Therefore, option (d) is the correct answer.

Knowledge Box
Siang River: ● The Siang originates near Mount Kailash in Tibet, where it goes by the name of Tsangpo. It traverses more than 1,000 km eastward, before forming a horseshoe bend around the towering Namcha Barwa peak, and enters Arunachal Pradesh as the Siang. Further downstream, in Assam, the river becomes the mighty Brahmaputra.

Relevance: The Adi tribe of Arunachal Pradesh are protesting against the proposed 11.2GW hydropower plant over the Upper Siang river.

Q14.
Answer: c
Explanation:

- **Alzheimer's** is **a progressive neurodegenerative disorder that primarily affects the brain's neurons**, leading to cognitive decline, memory loss, and behavioral changes. It gradually damages nerve cells, reducing brain function over time. Since it leads to the deterioration of the nervous system, it is classified as a degenerative disease. **So, statement 1 is correct.**
- **A key characteristic of Alzheimer's disease is the abnormal accumulation of amyloid beta (Aβ) proteins in the brain as plaques** (all the organs of the body are said to be placed in the body cavity only). These plaques disrupt neural communication, leading to cell death and memory loss. Alongside amyloid plaques, *tau* protein tanglement further contributes to neuron degeneration. **So, statement 2 is correct.**
- Though, Alzeihmer's is taken to as an age related degenerative disease, genetic link is often underscored as a contributing factor. There are two types of genes associated with the disease: risk genes and deterministic genes. Risk genes, like APOE-e4, increase the likelihood of developing Alzheimer's but do not guarantee it. In contrast, deterministic genes (APP, PSEN1, PSEN2) directly cause familial early-onset Alzheimer's, which accounts for less than 1% of all cases. Research has shown that **individuals with first-degree relatives (parents or siblings)** with Alzheimer's **have a higher chance of developing the disease** compared to those without a family history. **So, statement 3 is correct.**

So, all three of the above given statements are correct.
Therefore, option (c) is the correct answer.
Relevance: A new study of Ayurveda provides a multipronged approach to treatment of several neurodegenerative diseases, offers hope for Alzheimer's Disease, Dementia, and related diseases.

Q15.
Answer: a
Explanation:

- The term **orange economy** was first coined by former Colombian President Iván Duque Márquez. **It refers to businesses based on creativity, involving things like art, culture, content creation, film making, and social media influence.** It is a production model where goods and services have intellectual value because they are the product of the ideas and expertise of their creators. **That's why the orange economy is also known as the creative economy.**
- It is an evolving concept based on creative assets. Essentially it is a **knowledge-based economy** upon which 'creative industries' are based. Creative industries use creativity and intellectual capital as primary inputs to contribute to economic growth and development. UNCTAD and UNDP classify them by their roles as heritage, art, media and functional creations.
- These industries include **advertising, architecture, arts and crafts, design, fashion, film, video, photography, music, performing arts, publishing, research and development, software, computer games, electronic publishing and TV/radio.**

- The creative economy, also known as the orange economy, is one of the rapidly growing sectors of the global economy. The creative economy generates annual revenues of over $2 trillion and accounts for nearly 50 million jobs worldwide. As per a 2023 Global Innovation Report, India's creative economy is worth $30 billion. Also, 8% of India's total workforce is employed in creative occupations.

Therefore, option (a) is the correct answer.

Q16.
Answer: b
Explanation:

- **Thiruvalluvar**, was a Tamil poet and philosopher, much revered in Tamil Nadu. His historiography is debated with no conclusive assertion over his religion and period he belonged to.
- ***Tolkappiyam* is the earliest known work on Tamil grammar and linguistics**, but it was not written by Thiruvalluvar. It is **attributed to Tolkappiyar** and predates Thiruvalluvar's work. Thiruvalluvar is best known for composing the ***Thirukkural***, a classic Tamil text on ethics, morality, and governance. **So, statement 1 is not correct.**
- Thiruvalluvar strongly advocated for non-violence (*ahimsa*) and compassion, which aligns with the principle of vegetarianism. **His verses in the Tirukkural condemn killing and meat consumption, emphasizing kindness toward all living beings.** This is consistent with the broader ethical teachings found in Jainism and certain Hindu traditions. **So, statement 2 is correct.**
- Though his religious affiliations are debated along with the secularity and religiosity of the texts written by him, one stated fact is in his writings is he **refers to a divine energy and often praises the same**, this divinity is termed with ***Kadavul Vazhthu*** in his writings. **So, statement 3 is correct.**

So, only two of the above statements are correct.
Therefore, option (b) is the correct answer.

Q17.
Answer: c
Explanation:

- The **US Entity List** is a critical **trade control mechanism** maintained by the Bureau of Industry and Security (BIS) under the US Department of Commerce. It identifies foreign organizations, businesses, and individuals that are **subject to trade restrictions** because of national security, foreign policy concerns, or suspected involvement in activities that threaten US interests. The list is designed to prevent the transfer of sensitive technology, including advanced semiconductors, military equipment, artificial intelligence tools, and cybersecurity systems, to entities that may misuse them or threaten global stability.
- The primary objective of the US Entity List is to **regulate and restrict exports, re-exports, and transfers** (both within and outside the US) of certain **dual-use** and **controlled items** to listed entities. It ensures that US-origin technology and goods are not used for military, nuclear proliferation, cyber espionage, or human rights violations.
- Companies and organizations that are included in the list must seek a special export license from the US government before receiving any controlled items from US-based firms.

Therefore, option (c) is the correct answer.
Relevance: Recently, the US Government removed Bhabha Atomic Research Centre (BARC), Indira Gandhi Atomic Research Centre (IGCAR) and Indian Rare Earths Limited (IREL) from the US entity list.

Q18.
Answer: b
Explanation:

- **Etikoppaka toys** are handcrafted wooden toys originating from the Etikoppaka village in Andhra Pradesh. These toys are known for their eco-friendly nature and intricate craftsmanship.
- These toys were granted a GI tag in 2017 for **Andhra Pradesh** (not Karnataka). . **So, statement 1 is not correct.**
    - ***Channapatna toys*** *are GI tagged toys from Karnataka*.
- These toys are made to undergo a **lacquering process** that involves natural dyes and a friction-based polishing technique. The lacquer, derived from natural sources like seeds and tree bark, gives the toys their distinctive shine and smooth texture. This method has been passed down for generations, ensuring the toys retain their aesthetic appeal while remaining non-toxic. Because of this unique polish, these toys are often called **lacquer toys. So, statement 2 is correct.**
- **Ankudu wood (*Wrightia tinctoria*), a lightweight and soft material, is commonly used to carve these toys.** The artisans meticulously shape them to ensure there are **no sharp edges**, making them safe for children. The traditional crafting process emphasizes durability and smooth finishing, ensuring that the toys are both aesthetically pleasing and child-friendly. **So, statement 3 is correct.**
- Etikoppaka toys are coloured using natural dyes made from vegetable extracts, tree bark, and seeds. The vibrant hues come from organic sources such as turmeric for yellow, indigo for blue, and tamarind seed extract for darker shades.

So, only two of the above statements are correct.
Therefore, option (b) is the correct answer.
Relevance: The invitation by the President for the At-Home ceremony prominently featured Etikoppaka toys, highlighting their cultural and traditional significance.

Q19.
Answer: c
Explanation:

- The Government of India recently unveiled the '**SANJAY - The Battlefield Surveillance System (BSS).**'
- SANJAY is an automated system which integrates the inputs from all ground and aerial battlefield sensors. It then processes them and produces a Common Surveillance Picture of the battlefield. It will **enhance battlefield transparency and transform the future battlefield** through a Centralised Web Application which will provide inputs to Command & Army Headquarters, and the Indian Army Decision Support System.
- The SANJAY BSS is equipped with state-of-the-art sensors and cutting-edge analytics. It will monitor the vast land borders and will prevent intrusions. It will enhance Intelligence, Surveillance & Reconnaissance.
- SANJAY has been **indigenously & jointly developed by the Indian Army and Bharat Electronics Limited (BEL).**
- **Other Options:**
    - **DURGA II:** It is a laser based Direct Energy Weapon developed by DRDO.
    - **ASMI:** It is a machine pistol indigenously developed for the army.
    - **HAL RUDRA:** It is India's first indigenously developed attack helicopter developed by Hindustan Aeronautical Limited.

Therefore, option (c) is the correct answer.

Q20.
Answer: c
Explanation:

- **'Doctrine of essentiality'** was devised by the Supreme Court in the *Shirur Mutt case, 1954*. As per the doctrine only those practices can be extended protection under Article 25 of the constitution which are considered essential to the religion. These practices were thereon called the **Essential Religious Practices (ERP),** it constitute such practices which are considered to be at the core of the religious identity of a denomination.
- As to what constitutes as essential the Supreme court in the case, *Sardar Syedna Taher Saifuddin Saheb v. State of Bombay (1962),* addressed the issue of religious freedom and held **that the freedom to manage religious affairs includes the right to determine the essential practices of a religious denomination.** The court emphasized that the state should not interfere in matters of religious faith unless such practices are considered immoral or contrary to public order. **So, statement 1 is correct.**
- The determination of what constitutes an essential religious practice is primarily based on the tenets and doctrines of that religion, as interpreted by the community itself.
- It must be noted here that, though the denominations are well within their right to determine what is essential for their religious practice according to them, it **is for the courts to finally decide and uphold whether the practice is essential or not,** that is, the courts can overrule the classification.
- Article 25 of the Constitution guarantees the "freedom of conscience and the right freely to profess, practice and propagate religion". **However, this right isn't absolute and is subject to public order, morality, health, and other fundamental rights**. **So, statement 2 is not correct.**

So, Statement–I is correct, but Statement–II is incorrect.
Therefore, option (c) is the correct answer.
Relevance: Recently, the Bombay High Court ruled that the use of loudspeakers is not an essential religious practice, reaffirming the Supreme Court's stance that religious practices must be integral and indispensable to the faith to receive constitutional protection.

Q21.
Answer: d
Explanation:

- Apples are known for their cultivation requirement of high altitudes and near chilling temperatures. **HRMN-99** variety developed by Hariman Sharma is a **self- pollinating, low chilling variety of Apple.**
- HRMN-99 thrives in tropical, subtropical, and plain regions with summer temperatures. It thus enables apple farming in areas where it was previously considered unviable.
- The HRMN-99 variety is characterized by its striped, red-over-yellow skin, soft and juicy pulp, and ability to produce up to 75 kg of fruit per plant annually.

Therefore, option (d) is the correct answer.
Relevance: Shri Hariman Sharma, the developer of HRMN – 99 has been honored with Padma Shri recently.

Q22.
Answer: d
Explanation:

- **River frothing** is an occurrence characterized by the accumulation of foam on the surface of rivers. Frothing is linked to "excessive organic loads" entering the river system due to untreated sewage and industrial waste.

- Foaming or frothing is the product from interaction among **gas bubble** (generated by aeration, mechanical mixing and biological processes like denitrification and anaerobic digestion), **surfactants** (which come from the wastewater streams that contain slowly biodegradable surfactants, which reduce the surface tension of a liquid when dissolved), and **hydrophobic particles** (usually **filamentous bacteria with long-chain structures and a hydrophobic surface**).
- While foaming does occur at various points throughout the year, it tends to be **more prevalent in Indian rivers during the post-monsoon months of October and November.** It is due to two main factors: the river being in its lean phase and the flow of the water being low. These factors prevent the pollutants from getting diluted, and their accumulation brings visible froth to the water surface. **So, statement 1 is correct.**
- Another important factor is the **growth of filamentous bacteria** in the river. These types of bacteria most commonly are part of **human excreta containing sewage.** These bacteria have thread-like structures which can readily trap air bubbles, thus contributing to the formation of a stable foam on the water surface. **So, statement 2 is correct.**
- Further it is noted that streams with hydropower plants report higher incidences of frothing as the **turbulence caused by the release of water from these dams can also lead to froth formation,** especially during heavy rainfall periods. **So, statement 3 is correct.**
- **Foam reduces dissolved oxygen levels in the water** as most of the oxygen gets wasted in froth formation. It can thus lead to hypoxic conditions leading to the death of fish and marine life. **So, statement 4 is correct.**
    - Hypoxic condition means when a region, or a body is deprived of adequate levels of oxygen.
- The foam also presents risks of waterborne diseases for nearby communities that rely on groundwater linked to the river system.

So, all four of the above statements are correct.
Therefore, option (d) is the correct answer.

Knowledge Box
• Froth formation in rivers often indicates the presence of pollutants. One of the principal culprits behind froth is surfactants, substances commonly found in soaps and detergents. • These surfactants enter water systems primarily through untreated sewage and reduce the surface tension of water, facilitating the formation and persistence of bubbles. As rivers move and flow, these bubbles conglomerate into larger patches of foam. • Beyond surfactants, rivers also absorb significant amounts of organic matter. This includes decaying plant materials, dead organisms, and agricultural effluents. As these organic substances break down, they release gases. In surfactant-laden waters, these gases become trapped, further contributing to froth formation. The accumulation of such organic matter is often a result of agricultural runoff and poor waste management practices.

Relevance: The National Green Tribunal has sought a response from the Centre's Ministry of Environment, Forest and Climate Change and others over the presence of toxic foam in river Pennai in Tamil Nadu.

Q23.
Answer: b
Explanation:

- The **Geographical Indication (GI) tag** is a mark of authentication that assigns products to their precise geographical location. **A GI tag is conferred upon products originating from a specific geographical region**, signifying unique characteristics and qualities. GI tags are recognized under the Goods (Registration and Protection) Act, 1999 and are also recognized by the TRIPS agreement of the World Trade Organization.
- Rice varieties with GI Tag in India:
  - ***Karen Musley***: It is an indigenous rice variety from **Andaman and Nicobar Island**. It has been acclaimed for its aromatic flavour, rich nutritional content, and resilience in saline soils. The Karen community has cultivated it in a traditional way representing their cultural heritage and sustainable farming practices. It received the GI Tag in December 2024. **So, pair 1 is not correctly matched.**
  - ***Boka Saul*** or **'mud rice':** It is a peculiar kind of indigenous rice variety from **Assam.** This variety requires no cooking and needs to be soaked in cold water for about an hour to get it ready for eating. It received the GI Tag in 2018. **So, pair 2 is correctly matched.**
  - ***Kalanamak***: It is grown in **Terai region of Uttar Pradesh**. It is a traditional variety of paddy with a black husk, known for its extraordinary fragrance and health benefits. It is considered a 'gift from Lord Buddha' to the people, and it has been in cultivation since the Buddhist period (600 BC). It was awarded the Geographical Indication (GI) tag in 2013. **So, pair 3 is not correctly matched.**
  - ***Chak-Hao:*** It is a variety of **black rice** from **Manipur**. Chak-Hao is a scented glutinous rice which has been in cultivation in Manipur for centuries. It is characterised by its special aroma. Chak-Hao has also been used by traditional medical practitioners as part of traditional medicine. It was awarded the Geographical Indication (GI) tag in 2020. **So, pair 4 is correctly matched.**
  - ***Pokkali:*** It is a unique saline tolerant rice variety. It is grown in Ernakulam, Thrissur, Alappuzha, Kannur and Kottayam districts of **Kerala**. It is known for its saltwater resistance. It is grown as a part of a system in which the farming alternates between rice and prawn. The fields are used alternately for rice farming and prawn cultivation. Pokkali was conferred the Geographical Indications tag in the year 2008. **So, pair 5 is correctly matched.**

So, only three of the above given pairs are correctly matched.
Therefore, option (b) is the correct answer.
Relevance: Karen Musley, indigenous rice variety from Andaman and Nicobar have recently been accorded the GI tag.

Q24.
Answer: d
Explanation:

- **Digital Public Infrastructure (DPI)** refers to foundational digital systems that support essential public services. DPI can be taken as a set of **technology building blocks** that drive innovation,

inclusion, and competition at scale, operating under open, transparent and participatory governance. The internet, powered by common protocols like HTTP, HTML, and SMTP, is a prime example of DPI.

- DPI creates a balance, identifying frameworks consisting of **frequently required services and components efficiently managed by the public authority**. It develops open protocols, shared platforms, and enabling policies to create an interoperable ecosystem. The DPI **does not provide the complete solution** but makes the job of solution providers easier by offering readymade components and services.
- **Some common DPIs by Government of India:**
    - **Entity Locker:** It is a Digital Public Infrastructure aimed at streamlining the management of business documents. It has been **developed by the National eGovernance Division (NeGD), under the Ministry of Electronics and Information Technology (MeitY)**. It is a secure, cloud-based solution that simplifies the storage, sharing, and verification of documents. It provides for management of documents for a wide range of entities, including large organisations, corporations, micro, small, and medium Enterprises (MSMEs), trusts, startups and societies. **So, point 1 is correct.**
    - **Unified Payment Interface (UPI)**: It is an instant payment system developed by the **National Payments Corporation of India (NPCI)**. UPI is built over the IMPS (Immediate Payment Service) infrastructure and allows the user to instantly transfer money between any two parties' bank accounts. **So, point 2 is correct.**
    - **CoWIN:** It is the digital platform to capture and track covid-19 vaccination program. Co-WIN was critical in COVID related health services, including vaccination tracking and contact tracing. **So, point 3 is correct.**
    - **Ayushman Bharat Digital Mission:** It connects the digital health solutions of hospitals across the country with each other. It aims to create a unique health ID for every citizen (Ayushman Bharat Health Account), and a digital healthcare professionals registry and a facilities registry. **So, point 4 is correct.**
    - **Agri Stack:** It is an initiative of the **Ministry of Agriculture & Farmers Welfare**. Agri Stack is a database of Indian farmers, which contains their details like landholding, GPS coordinates of plots and the crops grown on them. Agri Stack will have a Farmer Registry, a federated registry of all the farmers in the country, compiled by States. Each farmer will be assigned a unique Digital ID and a digitally verifiable credential. **So, point 5 is correct.**

So, all five of the above are part of India's Digital Public Infrastructure.
Therefore, option (d) is the correct answer.
Relevance: The Ministry of Electronics and Information Technology (MeitY) has recently launched Entity Locker as a DPI.

Q25.
Answer: c
Explanation:

- **Graphics Processing Unit (GPU)** is a specialized processor originally designed to render images and graphics efficiently for computer displays.
- In recent years, GPUs have evolved into powerful co-processors to Central Processing Units (CPUs). The arithmetic computations are completed quickly by a GPU, freeing up the CPU to conduct other

activities or tasks, making them **indispensable for tasks beyond graphics, such as scientific simulations, artificial intelligence, and machine learning.**

- Unlike CPUs, which are designed for sequential processing and handle tasks one at a time, **GPUs are designed for parallel processing.** GPUs have thousands of smaller cores that can perform simultaneous calculations, making them highly efficient for tasks like rendering graphics or training machine learning models. This parallel processing capability is key to their efficiency in handling large-scale computations. **So, statement 2 is correct.**
- **Matrix multiplication** is a critical operation in many AI and machine learning tasks, particularly in deep learning. **GPUs are specifically optimized for matrix operations and large-scale linear algebra computations, which is why they are widely used for training deep neural networks.** Their architecture, including tensor cores in advanced models, further enhances the speed and efficiency of matrix multiplications, making them ideal for AI and scientific computing. **So, statement 1 is correct.**

Therefore, option (c) is the correct answer.

Knowledge Box
IndiaAI Mission: ● The Government of India under its flagship IndiaAI Mission has allocated nearly Rupees 4.5K crores to procure high-performance GPUs to enhance AI research and supercomputing capabilities in India. The initiative aims to ensure India has access to the necessary computational power for advanced AI research, despite global challenges like U.S. export restrictions.

Relevance: The USA has classified countries into three tiers to restrict the export of computing technologies, including GPUs. India falls under Tier II, which imposes limits on the computing power that can be imported from American companies.

Q26.
Answer: d
Explanation:

- **Lezim is a traditional folk dance of Maharashtra,** characterized by its energetic movements and synchronized formations.
- The dance gets its name from a musical instrument called Lezim, which consists of a wooden stick with jingling cymbals attached.
- It is often performed during **Ganesh Chaturthi processions**, military drills, and cultural festivals. The dance is accompanied by percussion instruments without any string or wind instruments.

Therefore, option (d) is the correct answer.
Relevance: A controversy emerged on the portrayal of Chatrapati Sambhaji Maharaj performing Lezim dance in a Hindi movie.

Q27.
Answer: c
Explanation:

- **Organophosphates** are chemical compounds formed through the esterification process involving phosphoric acid and alcohol. **These chemicals serve as primary components in herbicides, pesticides, and insecticides** and have extensive application in manufacturing plastics and solvents.
- These compounds are **neurotoxic to the animal body**, and act by blocking the neuromuscular transmission in the body. Organophosphates are antagonistic blockers of an enzyme cholinesterase, which is responsible for the destruction of the protein Acetylcholine. **So, statement 1 is correct.**
- Acetylcholine is responsible for transmitting signals from nerves to muscles, thus causing contractions, post the transmission it is destroyed by the enzyme cholinesterase. If Acetylcholine is not destroyed, then the muscle may not relax and can lead to generalized spasms. Such spasms in the respiratory muscles can even lead to death.
- Organophosphates are essential pharmacological products as well; they are administered in cases of reduced Acetylcholine production to ensure normal muscle functioning.
- One such condition is the **autoimmune disorder Myasthenia Gravis**, where the receptors for Acetylcholine get destroyed and thus a prolonged action of the protein is required for normal function of muscles. **Organophosphates in this case block the cholinesterase enzyme for prolonged sustenance of Acetylcholine**. **So, statement 2 is correct.**
- In case of Organophosphate Poisoning **Atropine from the plant *Atropa belladona*** is a potent antidote. **So, statement 3 is correct.**

So, all three of the above statements are correct.
Therefore, option (c) is the correct answer.
Relevance: The mystery deaths in Rajouri district of Jammu & Kashmir, are being attributed to Organophosphate poisoning.

Q28.
Answer: c
Explanation:

- NITI Aayog has recently launched the "**Fiscal Health Index 2025**". The index focuses on the fiscal health of Indian states. It offers data-driven insights that will be leveraged for informed state-level policy interventions to improve overall fiscal governance.
- **The Fiscal Health Index (FHI) aims to shed light on the fiscal status at the sub-national level** and guide policy reforms for sustainable and resilient economic growth.
- This was the first edition of this index and is stated to be an annual report from here on. The index ranks States on the basis of a composite fiscal index, which is based on five major sub-indices. These sub-indices are **quality of expenditure, revenue mobilization, fiscal prudence, debt index, and debt sustainability**.
- The index divides states into four categories based on their rankings. From top to bottom, they are Achiever, Front Runner, Performer and Aspirational.
- **Odisha tops the ranking**, followed by Chhattisgarh and Goa.
- Achievers are the top rankers. The top five high-performing states are Odisha, Chhattisgarh, Goa, Jharkhand, and **Gujarat**.
- The bottom five are aspirational which are Kerala, West Bengal, Andhra Pradesh, and **Punjab**.
- Punjab lies at the bottom of the table for overall fiscal health, while the State of **Bihar lies at the sixth place from the bottom.**

So, the correct sequence of the ranking of the States as per their overall fiscal health in the descending order is Odisha - Gujarat - Bihar - Punjab.
Therefore, option (c) is the correct answer.

Q29.
Answer: c
Explanation:

- **Stampede** is the surge of individuals in a crowd, in response to a perceived danger or loss of physical space. It often disrupts the orderly movement of crowds resulting in irrational and dangerous movement for self-protection leading to injuries and fatalities.
- The most common cause of death during stampedes is **Traumatic Asphyxia**, which occurs due to external compression on the chest and upper abdomen, leading to suffocation. Other possible causes include myocardial infarction (heart attack), head injuries, and crushing injuries to internal organs. **So, statement 1 is correct.**
- The NDMA **has issued detailed guidelines for crowd management**, which focus on capacity planning, hazard risk analysis, real-time surveillance, and structural improvements to prevent stampedes at mass gatherings. The NDMA's guidelines emphasize risk assessment, crowd flow management, and exit planning to mitigate such incidents. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.
Relevance: Recent stampedes at religious gatherings, including the Maha Kumbh in Prayagraj and Tirupati temple, have raised concerns over crowd safety and disaster preparedness.

Q30.
Answer: c
Explanation:

- **Article 309 of the Constitution of India** empowers Parliament and State Legislatures to regulate recruitment and conditions of service for Central and State Government employees. Thus, legislatures can frame laws governing public services, making this statement correct. **So, statement 1 is correct.**
- Public employment in India is based on **Article 310** of the Constitution, also known as the **"Doctrine of Pleasure."** It states that **Central government employees hold office at the pleasure of the President, and State government employees serve at the pleasure of the Governor.** However, this power is **not absolute** and is subject to **constitutional safeguards** provided under **Article 311**, which ensures protection against arbitrary dismissal, removal, or reduction in rank. **So, statement 2 is correct.**
- Further, **Article 311(2) states that no public servant can be dismissed, removed, or reduced in rank without being given a reasonable opportunity to be heard**. This ensures procedural fairness in service matters. **So, statement 3 is correct.**

So, all three of the above statements are correct.
Therefore, option (c) is the correct answer.
Relevance: In the recent controversy regarding the lateral entry scheme of the Central Government, the Doctrine of Pleasure (DoP) under Article 310 and Article 309 is at loggerheads. As per interpretation of Article 309 recruitment to central jobs can only be through an act of parliament or a statutory rule made by the President, while it is countered by the argument that DoP enshrined in Article 310, provides for the President to appoint any person who have specific qualifications for the post.

Q31.
Answer: b
Explanation:

- **A rocket launch site** is a strategically located location to send probes into outer space, it is selected on the parameters of safety in case of any untoward incident related to the launcher as well as to gain maximum impetus from the geography of the site.
- **Most of the launch sites are located on the East coast of the continents**, this is because launching **eastward aligns with the Earth's rotation (West to East)**, helping satellites gain extra velocity, thereby improving fuel efficiency. This additional velocity, around 1670 km/h at the equator, reduces the amount of energy required to propel the satellite into orbit. **So, statement 2 is correct.**
- The other reason is safety, **launching from the eastern coast ensures that debris from the launch falls into the ocean** rather than populated areas, reducing risks in case of failures.
- As far as the eastward push is concerned, this benefit is largely concerned with satellites which are placed in geo-stationary orbit or which circle the Earth parallel to the equator. Such satellites are usually communication satellites or satellites used for scientific research such as the ISS.
- There are other satellites which are placed in **polar orbits** moving across the equator in north-south direction and used mainly for mapping (such as **Earth Observation satellites**). **Such satellites are generally launched in southward or northward direction and therefore cannot take advantage of the Earth's rotation. So, statement 1 is correct.**
    - Still they are launched from East Coast sites for the site's added **advantage of safety**.

So, both Statement–I and Statement–II are correct, but Statement–II does not explain Statement–I.
Therefore, option (b) is the correct answer.
Relevance: Union cabinet has given approval for establishment of a third launch site of India at Sriharikota.

Q32.
Answer: b
Explanation:

- Southern Indian peninsula, have reported many of the archaeological sites belonging to the prehistoric settlements, megalithic iron age and later periods.
- Some of the prominent archaeological sites in the region are:
    - **Adichanallur (Thamirabarani River)**: It is located in the Thoothukudi district, Tamil Nadu, and is largely attributed to the megalithic period in Tamil Nadu. This site is located on the banks of the **Thamirabarani river.** Peculiar Tamil-Brahmi inscriptions on the urns used for burials are discovered from this site. **So, pair 1 is not correctly matched.**
    - **Keezhadi (Vaigai River):** This site is located in the Sivaganga district of Tamil Nadu on the banks of **Vaigai river**. More than 15,000 artefacts have been discovered from this site, including terracotta figurines, spindle whorls, Tamil-Brahmi inscriptions, and iron tools. The site has reported advanced urban settlement with well-laid streets, ring wells, and evidence of literacy. The site is directly linked to Sangam literature, which extensively mentions the Vaigai River and its influence on civilization. **So, pair 2 is not correctly matched.**

- **Kodumanal (Noyyal River):** This site is located in the Erode district, Tamil Nadu, on the banks of **Noyyal River**. It was an industrial site known for iron smelting, jewelry-making, and trade with the Roman world. **So, pair 3 is correctly matched.**
- **Pattanam (Periyar River):** Located in Ernakulam district, Kerala, this site corresponds to the ancient Muziris Port (1st Century BCE – 5th Century CE). Evidence of Indo-Roman trade, including glass beads, semi-precious stones, and iron implements have been found at this site. **So, pair 4 is correctly matched.**

So, only two of the pairs given above are correctly matched.
Therefore, option (b) is the correct answer.
Relevance: Recent archaeological findings have given winds to the debate regarding chronology of Iron age in South India, which has also garnered political attention.

Q33.
Answer: b
Explanation:

- **The Annual Status of Education Report (ASER) 2024 was released by the NGO Pratham,** [*not by the Ministry of Education*]. Since 2005, it has been releasing the ASER to measure basic reading and arithmetic levels among school children, attendance in school and other indicators. The data have revealed broad trends in learning over the years. **So, statement 1 is not correct.**
- For the first time, ASER 2024 has included a **section on digital literacy** among children aged 15-16 years, evaluating their ability to search for information online, set alarms, and browse the internet. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.

Knowledge Box

Key Findings of ASER 2024 Survey:
• **Preschool Enrollment (Ages 3-5)**
  o **77.4%** of rural children aged **3 years** are enrolled in early childhood education programs (Anganwadi/LKG/UKG).
  o Significant improvement in **preschool coverage** between **2018 and 2024**.
• **Elementary Education (Ages 6-14)**
  • **Improvement in reading and arithmetic levels** across most states compared to the **2022 ASER report**.
  • Significant progress in **Classes 1-3**, recovering from **learning losses during COVID-19**.
  • Schools increasingly implement **Foundational Literacy and Numeracy (FLN) activities** under the **NIPUN Bharat** initiative.
• **Implementation of FLN (Foundational Literacy and Numeracy)**
  o **83% of schools** reported receiving government directives for FLN implementation.
  o **78% of schools** had at least one teacher trained in FLN techniques.
  o **75% of schools** received relevant **FLN learning materials**.
• **Early Childhood Care and Education (ECCE)**

- **Shift in Class 1 enrollment age** to **6 years**, aligning with the **National Education Policy (NEP) 2020**.
- Growing emphasis on **preparing children for formal schooling through Anganwadis and preschool programs**.
- **Himachal Pradesh and Punjab** shifting toward **pre-primary classes** in schools.
- **Rajasthan seeing increased enrollment of 5-year-olds** in both Anganwadis and private LKG/UKG.

- **Digital Literacy and Smartphone Access (Ages 15-16)**
  - **90% of rural adolescents** now have **access to a smartphone**.
  - **80.1% of boys** and **78.6% of girls** (ages 14-16) could perform basic **digital tasks** like browsing information online.
  - Some **southern states** reported girls performing equally or better than boys in **digital skills**.
- **Enrollment Trends for Older Children (Ages 15-16)**
  - **Decline in dropout rates**, with **only 7% of students in this age group not enrolled in school**.
  - Continued focus needed on ensuring **secondary education access and completion**.
- **Budget and Teacher Recruitment Needs for ECCE**
  - NEP 2020 recommends **dedicated teachers for early childhood education**.
  - Budget planning required for **recruiting, training, and sustaining ECCE teachers**.
  - Need for additional resources to **strengthen early childhood education in Anganwadis**.
- **Gender Gaps in Digital Literacy**
  - **Minor gender gap in digital access and skills** (boys slightly ahead).
  - In **some southern states**, **girls outperformed or matched boys** in digital competency.

Q34.
Answer: c
Explanation:

- A Public Interest Litigation (PIL) is a form of litigation that is filed in a court to safeguard or enforce public interest. In India, the concept of PIL was introduced in the 1980s as a product of the judicial activism of the Supreme Court of India.
- **PIL has not been defined in any Indian statute or the Constitution.** However, the Supreme Court of India has held that the expression 'PIL' means a legal action started in a court of law for the enforcement of public/general interest where the public or a particular class of the public has some interest (including pecuniary interest) that affects their legal rights or liabilities.

- PILs are **extensions of Writ Jurisdiction** of the Supreme court and the High Court. Therefore, PILs may be filed either before the Supreme Court of India under Article 32 of the Indian Constitution or any High Court under Article 226 of the Indian Constitution. **So, statement 1 is correct.**
- As doctrined and reiterated by the Supreme Court, the requirement of ***locus standi*,** that only the aggrieved party or the party having specific interest to the dispute can appear or appeal in the court of law**, doesn't fully apply in cases of PILs,** any person, whether party to the dispute or not can file a PIL in Indian courts, the only precondition being that it must prove to the satisfaction of the court that the petition is being filed for public interest. **So, statement 2 is correct.**
- ***Res Judicata* is a common law doctrine meant to bar re-litigation of cases between the same parties in Court.** It means that if a case between the parties has been directly or indirectly tried and decided in a court, it will not be tried again in the same or different court.
- The **principle of Res Judicata does not apply strictly to PILs.** As dictated by the Supreme Court, the principle of Res Judicata or any principles analogous to it would depend on the circumstances and facts of the case and the nature of the PIL. **So, statement 3 is not correct.**
    - The apex court held that the principle would apply to public interest litigation provided it was a bona fide litigation. So, the principle of res judicata is applicable, but not strictly.
- As under the Supreme Court rules (which are also applicable to the High Courts) under Order 38 Rule 12(1)(a) a ***suo motu* cognizance taken by the court is to be taken as a PIL**. **So, statement 4 is correct.**

So, only three of the above given statements are correct.
Therefore, option (c) is the correct answer.
Relevance: The Hon'ble Vice-President of India, recently remarked that the access to judiciary has been weaponized in recent years and it is posing a great challenge to our governance.

Q35.
Answer: b
Explanation:

- The **World Health Organization (WHO)** is a specialized agency of the United Nations (UN) responsible for global public health.
- WHO membership is **not limited to UN member states.** All countries which are members of the UN may become members of WHO by accepting its constitution. Other countries may be admitted as members when their application has been approved by a simple majority of the World Health Assembly. **So, statement 1 is not correct.**
- WHO's constitution allows a member country to **withdraw by giving a one-year notice** and settling any outstanding financial obligations. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.

Knowledge Box
• The **World Health Organization (WHO)**, established in 1948, is a specialized agency of the United Nations that connects nations, partners and people to promote health, keep the world safe and serve the vulnerable – so everyone, everywhere can attain the highest level of health.

- WHO is responsible for providing leadership on global health matters, shaping the health research agenda, setting norms and standards, articulating evidence-based policy options, providing technical support to countries and monitoring and assessing health trends.
- The governance of WHO operates through the **World Health Assembly**, which meets annually as the general policy-making body, and through an **Executive Board of health specialists** elected for three-year terms by the assembly.
- The agency is led by a director general nominated by the Executive Board and appointed by the World Health Assembly.

Relevance: The US administration has recently announced its decision to withdrew from the WHO.

Q36.
Answer: c
Explanation:

- **Neglected Tropical Diseases (NTDs)** are a group of infections that are most common among marginalized communities in the developing regions of Africa, Asia and the Americas.
- They are caused by variety of **pathogens such as viruses, bacteria, protozoa and parasitic worms**. NTDs are especially common in tropical areas where people do not have access to clean water or safe ways to dispose human waste. These diseases **generally receive less funding for research and treatment** than malaises like Tuberculosis, HIV-AIDS etc. hence are called neglected.
- The World Health Organization releases a list of NTDs, with January 30 being declared as the day for generating awareness regarding NTDs.
- Some NTDs which are prevalent in India:
    - **Lymphatic Filariasis:** It is commonly known as **elephantiasis**. Infection occurs when filarial parasites are transmitted to humans through **mosquitoes**. It is caused by infection with parasites classified as nematodes (roundworms) of the family Filariodidea. Infection is usually acquired in childhood and causes hidden damage to the lymphatic system. It damages the lymphatic system and can lead to the **abnormal enlargement of body parts,** causing pain, and severe disability. **So, point 1 is correct.**
    - **NOMA**: It is a severe gangrenous disease of the mouth & face. It has a mortality rate of approximately 90% amongst the affected individuals. It primarily affects children who are malnourished, living in extreme poverty with poor oral health and weakened immune systems. This disease is also called the **face of poverty** as effective drugs like sulfonamides, penicillin etc. remain inaccessible to many due to extreme poverty. It was included in the list of NTDs in 2023. **So, point 2 is correct.**
    - ***Kala-Azar*** or **Visceral Leishmaniasis:** It is caused by protozoan parasites which are transmitted by the bite of infected female phlebotomine sandflies. It can cause severe debility, malnutrition and loss of ability to work. **It is fatal if left untreated in over 95% of cases. So, point 3 is correct.**
    - **Dengue**: It is a viral infection which is transmitted by ***Aedes aegypti*** mosquitoes. Dengue is found in tropical and sub-tropical climates worldwide, mostly in urban and semi-urban areas.

Most people with the infection report mild symptoms, but the disease is known to cause fever, severe headache, muscle and joint pains, nausea and vomiting, pain behind the eyes, and rashes. In severe cases, however, the infection can lead to internal bleeding, and if not managed properly, even death. It is included in the list of NTDs by WHO. **So, point 5 is correct.**

So, only four of the above given points are correct.
Therefore, option (c) is the correct answer.
Relevance: On the World Neglected Tropical Diseases day 2025, the Government of India decided to light up the India gate in Delhi with vibrant hues of purple and orange

Q37.
Answer: b
Explanation:

- The Wetlands (Conservation and Management) Rules, 2017 empower both the **central and state governments** to notify wetlands. State governments have the authority to identify and notify wetlands within their respective jurisdictions. **So, statement 1 is not correct.**
- According to the Wetlands (Conservation and Management) Rules, 2017, **river channels are excluded from the definition of wetlands** in India. Other excluded water bodies include paddy fields and man-made reservoirs, constructed for specific purposes like irrigation and drinking water supply. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.
Relevance: Recently, the National Green Tribunal (NGT) has asked the Uttar Pradesh government to submit a status update on the Dhanauri wetland.

Q38.
Answer: b
Explanation:

- Biodiversity Heritage sites (BHS) are protected areas under the Biodiversity Act, 2002. Under this act the state government can notify the sites known for their unique biodiversity as BHS. These are well defined areas that are unique, ecologically fragile ecosystems - terrestrial, coastal and inland waters and, marine having rich biodiversity.
- The **Arittapatti Biodiversity Heritage Site** is a significant ecological and historical site located in Tamil Nadu, India. It is home to around 250 species of birds, including three important raptor (bird of prey) species. Notified in 2022, it is the first and **only biodiversity heritage site in the State of Tamil Nadu. So, statement 1 is correct.**
- The Arittapatti supports close to 72 lakes, 200 natural spring pools and three check dams. Among other lakes, the Site includes the famous 16th century lake **Anaikondan lake**, it is an important reservoir supporting a large number of waterfowls, but is **not included in the Ramsar list** of wetlands. **So, statement 2 is not correct.**
- Arittapatti Biodiversity Heritage Site is home to 250 species of birds including three flagship raptor species – Laggar Falcon (*Falco jugger*), Shaheen Falcon (*Falco peregrinus*) and Bonelli's Eagle (*Aquila fasciata*). It is also home to wildlife such as the Indian Pangolin, **Red Slender Loris** and pythons. **So, statement 3 is correct.**
    - Slender Loris is a small nocturnal mammal that spends most of its life on trees. They are native to India and Sri Lanka and are also amongst the world's top endangered primates. They are listed as **endangered** in the International Union for the Conservation of Nature (IUCN) red

list. They are also protected under Schedule I of the Indian legislation of the Wildlife (Protection) Act (WPA), 1972.

So, only two of the above given statements are correct.
Therefore, option (b) is the correct answer.
Relevance: Mines Ministry decided to Annul the Auction of Nayakkarpatti Tungsten Mineral Block, which was located inside the Arittapati biodiversity heritage site.

Q39.
Answer: b
Explanation:

- Bail can be described as a security such as money or bond, required by a court for the release of a prisoner, who must appear at a further date. The term *per se* is not defined anywhere in the Indian criminal codes, but specific provisions in the same are provided for the release of prisoners.
- Types of bails that can be granted in India:
  - **Regular Bail:** It is **granted after a person has been arrested and taken into custody**, allowing them to be released during the trial. It ensures that the accused does not have to remain detained while their case is being heard, provided they comply with court conditions. The primary objective is to uphold the right to personal liberty while ensuring the accused's presence during legal proceedings. **This type of bail is granted while the investigation is still ongoing or impending**. **So, pair 1 is not correctly matched.**
  - **Anticipatory Bail:** This kind of bail is granted in anticipation of arrest, **allowing the accused to avoid detention before they are actually taken into custody.** Once granted, it prevents immediate arrest but may have conditions attached, such as cooperating with the investigation. **So, pair 2 is correctly matched.**
  - **Interim Bail:** It is a temporary bail granted before the hearing for regular or anticipatory bail. **It provides short-term relief to an accused person until the final decision is made on their main bail application**. Interim bail can be extended but, if not confirmed, the accused must surrender or risk arrest. This type of bail is granted to ensure that the accused is not unfairly detained while their bail application is still pending. **So, pair 3 is correctly matched.**
  - **Statutory Bail:** This type of bail is also known as "default bail" and is provided if the investigation is not completed within 60 or 90 days (depending on the offense), **the accused is entitled to bail as a matter of right.** This provision prevents the police from keeping individuals in custody indefinitely without filing charges against them. **So, pair 4 is not correctly matched.**

So, only two of the above given pairs are correctly matched.
Therefore, option (b) is the correct answer.

Knowledge Box
Transitory Bail: ● Apart from the bails mentioned above, there is another type of bail – the **Transitory bail** – The Supreme Court introduced this concept in 1998. Transit application bail serves as a form of protection for the accused against arrest until they can reach a court with territorial jurisdiction for the alleged offense. This

type of bail **provides equitable and interim relief**, particularly for **individuals residing in a different state**, allowing them to seek anticipatory bail.

Relevance: Recently Supreme Court have raised concerns regarding the lackluster approach of HCs in granting bail

Q40.
Answer: c
Explanation:

- The act of attempting to suicide under the Indian Penal Code (IPC) was a punishable offence, it was later decriminalised by the promulgation of Mental Health Act (MHA) of 2017, under the current Bharatiya Nyay Sanhita, 2023, attempt to suicide is not mentioned as an act to attract punishment.
- The BNS, 2023, which replaces the Indian Penal Code (IPC), have **decriminalized the act of attempting suicide**. However, an exception is made for cases where the attempt is used as a means to obstruct public officials in their duties, which may still invite legal consequences. This reflects a progressive shift in India's criminal law framework, recognizing the mental health dimension of suicide attempts rather than treating them as criminal acts. **So, statement 1 is correct.**
- **Mental Health Act, 2017 overrides Section 309 of the IPC** (section related to suicide), acknowledging that a person who attempts suicide is likely to be under severe mental distress. It mandates that such individuals receive appropriate medical and psychiatric care rather than being subjected to punitive measures. The law further directs the government to ensure rehabilitation and support systems to reduce the risk of recurrence, thereby shifting the focus from punishment to mental health care and prevention strategies. **So, statement 2 is correct.**
- It is to be noted here that attempt to suicide was and to this date is a criminal offence under the IPC, the MHA just overrides the related provisions.

Therefore, option (c) is the correct answer.
Relevance: The Supreme Court recently delivered a significant ruling on abetment to suicide, reiterating that it remains a punishable offense under Sections 106 and 105 of the Bharatiya Nyaya Sanhita (BNS), 2023.

Q41.
Answer: c
Explanation:

- Dengue is a tropical disease caused by a virus belonging to the flaviviridae family. It is transmitted through mosquitoes of the species *Aedes Aegypti*, which is peculiar for its preference of clear water for breeding.
- Recent studies by the Indian Institute of Tropical Meteorology (IITM) in Pune found that **the incidence of dengue increases during wet weeks with moderate rainfall but decreases with extreme heavy rainfall.** The study also highlighted the importance of early warning systems based on meteorological conditions.
- Research indicates that **very heavy rainfall (above 150 mm) can flush out mosquito larvae and eggs from breeding sites,** reducing the risk of dengue transmission. However, moderate rainfall (0.5 mm to 150 mm) creates stagnant water pools, increasing mosquito breeding and dengue cases. **So, statement 1 is correct.**
- *Aedes aegypti*, the primary vector for dengue, **does not breed in muddy water, swamps, or creeks. Instead, it prefers clean, stagnant water** found in artificial containers such as discarded

tires, flowerpots, water storage tanks, and open drains in urban and semi-urban areas. **So, statement 2 is not correct.**

So, Statement-I is correct, but Statement-II is not correct.
Therefore, option (c) is the correct answer.

Q42.
Answer: c
Explanation:

- India and Indonesia are joint partners in the Indian Ocean, having economic, cultural and strategic ties. These two are partners in the East Asia Summit, Indian Ocean Rim Association etc.
- **East Asia Summit (EAS):** It is a forum for strategic dialogue involving **ASEAN+8 countries**, including **both India and Indonesia**. EAS includes 10 ASEAN countries alongwith Australia, China, India, Japan, New Zealand, Russia, South Korea, and the United States. **So, point 4 is correct.**
- Other options:
  - **ASEAN+3:** It is a forum that includes the **10 ASEAN nations** (including Indonesia) plus **China, Japan, and South Korea** to enhance regional cooperation. India is NOT a part of ASEAN+3. **So, point 1 is not correct.**
  - **BIMSTEC (Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation):** It is a regional organization linking **South Asia and Southeast Asia**. BIMSTEC includes Bangladesh, **India**, Myanmar, Sri Lanka, Thailand, Nepal, and Bhutan. Indonesia is NOT a member of BIMSTEC. **So, point 2 is not correct.**
  - **Ganga-Mekong Cooperation:** A cultural and economic cooperation initiative between **India and five east Asian countries lying in the Mekong basin.** It is often referred to as India + 5 (Cambodia, Laos, Myanmar, Thailand, Vietnam) grouping. Indonesia is NOT a part of Ganga-Mekong Cooperation. **So, point 3 is not correct.**
  - **Regional Comprehensive Economic Partnership (RCEP):** It is a **major trade agreement** between ASEAN and its FTA partners RCEP originally included both Indonesia and India during negotiations. However, India opted out of RCEP in 2019 due to concerns over trade imbalances and protection of domestic industries. **So, point 5 is not correct.**

So, only four of the above groupings do not include both India and Indonesia together as partners
Therefore, option (c) is the correct answer.
Relevance: In 2025, Indonesian President Prabowo Subianto was invited as the chief guest for India's Republic Day, emphasizing deepening maritime security and economic cooperation between the two countries.

Q43.
Answer: a
Explanation:

- The **Human Development Report 2024** released by the United Nations underscored the recent trends in democratic politics, where people are preferring and supporting the democratic systems, but at the same time are electing and supporting those leaders who have been often known to have or have leanings to undermine the democratic elements. This situation has been termed as **Democracy Paradox**.
- Other Options:

- **Dog Whistle Politics:** The practice of conveying an insidious implicit message to a certain set of targeted audience, covertly through means of public harmless looking actions or speeches for political gains
- **Gerrymandering:** The practice of drawing electoral boundaries or constituencies in such a way that gives one party an unfair advantage over its rivals.
- **Heckler's Veto:** A person or a group of persons who are not in agreement with a lawful procedure chose to obstruct it via heckling (creating nuisance).

Therefore, option (a) is the correct answer.

Q44.
Answer: b
Explanation:

- **Vadnagar town is a multi-layered and multi-cultural mercantile settlement with its history stretching back to nearly 8th Century BCE**. Vadnagar is one such historic town which has reported continuous habitation for more than 2700 years.
- **Ahmedabad and Jaipur are the two Indian cities in the UNESCO World Heritage Sites**. Vadnagar is in the tentative list of World Heritage Sites, on recommendations of the Government of India. **So, statement 1 is not correct.**
  - A Tentative List is an inventory of those properties which each State Party intends to consider for nomination.
- **Sharmishtha Lake is a large body of water near Vadnagar.** The ancient town of Vadnagar is an L-shaped town with Sharmishtha Lake located on its northeastern edge. **So, statement 2 is correct.**
  - The Sharmishtha Lake is an interesting example of the water retaining systems of the Solanki period, as it is still in use. It is an ancient lake created by the waters of the river Kapila flowing from the hills of Aravalli.

Therefore, option (b) is the correct answer.
Relevance: The Union Home Minister recently inaugurated the state-of-the-art Archaeological Experiential Museum in Vadnagar, Gujarat. The four-storey museum covers over 2,500 years of human evolution.

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Q45.
Answer: c
Explanation:

- Ketorolac, Acecloflenac, Dicloflenac, and Nimesulide belong to a group of drugs known as **Non-Steroidal Anti-inflammatory Drugs (NSAIDs)**, these drugs are prescribed for relieving pains, inflammation and are also known for their antipyretic properties (fever relieving).
- The above-mentioned four drugs have been under scrutiny in recent years because of their **toxicity to vultures**. These NSAIDs are used on large scale in veterinary care, many a times unregulated. Improper disposal of animal carcasses treated with such a drug lead the carcass to be fed upon by scavengers like vultures, crows, dogs etc. These drugs are nephrotoxic to vultures (toxic to the renal system of vultures), leading to birds' death.

- Earlier, **diclofenac was one of the prime reasons for wiping out 99 percent of the vulture population in India in the 1990s.** The use of diclofenac has been banned in India for veterinary purposes. Now **Nimesulide has also been banned for production and distribution** in the country for the same reason.
- Two more veterinary drugs – **aceclofenac** and **ketoprofen** – are also found to be toxic for vultures. A proposal to ban these two drugs is under consideration by the Indian government.
- Vultures are essential for maintaining ecosystems. They are often referred to as nature's sanitation workers. They play a crucial role in maintaining the ecosystem health by swiftly removing dead animals that might otherwise spread harmful diseases.

Therefore, option (c) is the correct answer.

Relevance: The Government of India recently banned the manufacture, sale and distribution of all formulations of Nimesulide because of its toxicity towards vultures.

Q46.

Answer: c

Explanation:

- The community forest resource area is the common forest land that has been traditionally protected and conserved for sustainable use by a particular community.
    - The community uses it to access resources available within the traditional and customary boundary of the village; and for seasonal use of landscape in the case of pastoralist communities.
- According to the Section 2(a) of the **Forest Rights Act, 2006** a "community forest resources" is the "customary common forest land within the **traditional or customary boundaries of the village**, including reserved forests, protected forests and protected areas such as sanctuaries and national parks to which the community had traditional access". Thus, no defined boundaries are there for the Community resource region. **So, statement 1 is correct.**
- Section 3(1)(i) of the FRA is referred to as the Community Forest Resource Rights (CFRR). They recognise not just forest communities' rights to access and use forest produce, but also their "rights to protect, regenerate or conserve or manage any community forest resource which they have been traditionally protecting and conserving for sustainable use".
- **Once CFRR is recognised for a community, the management of the forest passes into the hands of the gram sabha, instead of the forest department**. The gram sabha is free to form its own rules and regulations for management, use and conservation.
- The **Gram Sabha thus get the statutory authority** to protect, regenerate, conserve or manage community forest resources, along with the wildlife, flora, and biodiversity within. The gram sabhas would also have been responsible for preventing activities that harm their cultural and natural heritage. Thus, community forest resource areas come under the jurisdiction of gram sabhas. **So, statement 2 is correct.**
    - The Gram sabha would have to constitute its own Community Forest Resource Management Committees to develop and execute the managed plan approved by the gram sabha. And State governments would be duty-bound to support these plans.

Therefore, option (c) is the correct answer.

Knowledge Box
Sacred Grove:

- Sacred groves are community-regulated and conserved patches of forest land.
- They are created and managed traditionally by communities in various ways deeply rooted in their identity.
- The groves are kept inviolate through customary laws and taboos. Many of these customary laws completely prohibit the extraction of any resources in any form, except for custodians to access medicinal plants.
- The grove's conservation ethos is strongly linked to piety and sanctity, enforced in terms of the community's relationship with its spirits and gods.
- India is estimated to have 1-10 lakh sacred groves of this nature, the highest in the world.
- They are often associated with temples, monasteries, shrines, pilgrim sites, and/or burial grounds.
- Local legends, myths, and beliefs surround sacred groves. Communities have transmitted them orally through several generations as a matter of conserving their cultural identities.

Relevance: The Supreme Court recently asked the Rajasthan Government to identify the traditional communities historically protecting the sacred grooves and designate the areas as Community Forest Resource under FRA.

Q47.
Answer: b
Explanation:

- **Kerb trading** refers to unofficial or after-hours trading that occurs outside formal stock exchanges, often on the streets or in informal settings. It was historically used when exchanges closed early or had restrictions on trading certain stocks. While less common today due to electronic trading, it still exists in some markets where regulations limit trading hours. This type of trading is often less regulated and riskier than official market transactions.
- Other Options:
  - **Scalping**: It is a day trading strategy where traders execute multiple small trades to profit from minute price changes. It relies on high trade volumes and rapid executions, often within seconds or minutes, to generate profits. Scalping is conducted on regulated exchanges and does not involve informal or after-hours trading like kerb trading. Therefore, it is unrelated to kerb trading.
  - **Short selling**: It involves selling borrowed shares in anticipation of a price drop, allowing traders to buy them back later at a lower price for a profit. It is a legitimate market strategy regulated by stock exchanges and is used to hedge risks or speculate on declining stocks. Unlike kerb trading, short selling occurs within formal trading hours and under strict market regulations.
  - **Greenshoe option**: It is a financial tool used in initial public offerings (IPOs) where underwriters sell extra shares (up to 15% more than originally offered) to stabilize stock prices if demand is high. This option helps prevent extreme price volatility in newly listed stocks. Since kerb trading involves informal market activities, it has no connection to the greenshoe mechanism, which operates under strict regulatory oversight.

Therefore, option (b) is the correct answer.

Relevance: The Securities and Exchange Board of India (SEBI) is planning to introduce a "when-listed" platform, which will allow trading companies' shares in a period between allotment of shares post the closure of the initial public offering (IPO) bidding process and the official listing on stock exchanges. It is a move to curb Kerb trading.

Q48.
Answer: b
Explanation:

- Dracunculiasis or Guinea-worm disease is an **endoparasitic disease** (diseases in which the parasite lives inside the body of the host are called endoparasitic diseases) caused a **helminthic organism** [organisms belonging to class Platyhelminthes (flatworms such as TapeWorm) and Aschelminthes (roundworms) are commonly referred to as helminthes)]. **So, statement 1 is not correct.**
- It is classified as a **Neglected Tropical Disease by the WHO** and presents as painful blisters on the skin. It results from ingesting the parasite *Dracunculus medinensis* or Guinea-worm. Transmission mostly occurs through drinking stagnant water that is contaminated by parasite-infected water fleas.
- *Dracunculus medinensis* is the **largest of the tissue parasites known to affect humans**. **So, statement 2 is correct.**
- **Domesticated dogs and cats can act as carriers for these diseases**, the ingestion of infected water by these animals leads to water fleas entering their body, the flea itself gets destroyed while the larvae can mitigate and pass through urine and stools of the organism. The carriage of the disease by these domesticated animals is a major obstacle in the eradication of the disease. **So, statement 3 is correct.**
- The Guinea-worm disease has been eradicated from India. Certification of India as a Guinea Worm disease free country by the World Health Organisation was granted in February 2000.
    - Guinea worm is the second communicable disease after smallpox, that has been eradicated from the country.

So, only two of the above given statements are correct.
Therefore, option (b) is the correct answer.
Relevance: Former US President Jimmy Carter, who recently passed away was known for his contribution regarding this Neglected Tropical Disease (NTD).

Q49.
Answer: a
Explanation:
Rwanda is a landlocked country in the Great Rift Valley of East Africa. It shares land borders with **Uganda, Tanzania, Burundi, and the Democratic Republic of Congo.**

So, only two of the above given points are correct.
Therefore, option (a) is the correct answer.
Relevance: Newly appointed US President, soon after assuming the office have blamed Rwanda for supporting the M23 rebel group against its neighbor Democratic Republic of Congo

Q50.
Answer: b
Explanation:

- **Plea Bargaining** is an agreement between defense and prosecution of a court case, where the **accused pleads guilty in lieu or expectation of a lesser offense or a reduced sentence.**
- Plea bargaining was recognized under the erstwhile Criminal Procedural Code, 1973 as well as is included in the Bharatiya Nagrik Suraksha Sanhita (BNSS), 2023.
- Other Options:
    - **Forum Shopping:** Practice of litigants deliberately filing cases at a specific court, in the hope of getting a favourable outcome.
    - **Black-court syndrome:** Anxiety experienced by the ordinary citizen in court settings.
    - **Public Interest Immunity:** A legal doctrine that allows the State to withhold sensitive information from being disclosed in court proceedings.

Therefore, option (b) is the correct answer.
Relevance: As per recent data by the Ministry of Law and Justice only 0.11% of all criminal cases have been disposed of via plea bargaining.

Q51.
Answer: d
Explanation:

- Solar geoengineering is a set of techniques aimed at cooling the Earth by reflecting sunlight or reducing its absorption. These methods include **Aerosol Injection**, **Marine Cloud Brightening**, **High-Albedo Crops and Buildings**, **Ocean Mirror**, and **Cloud Thinning.** While they hold potential for mitigating global warming, these methods also raise significant environmental, ethical, and technical challenges.
- As per a recent report **'No-till' farming,** in which seeds are sown into fields **without first ploughing** them, could help to lower temperatures near croplands by up to 2 °C. The effect is driven by the increased fraction of sunlight that the soil reflects back into space, or albedo, in fields that have not been ploughed, which reduces the amount of heat the Earth's surface absorbs from the Sun. No-till plots are roughly 50% more reflective during summer months than tilled croplands, according to measurements made in the wheat fields of an experimental farm site in the Provence region of France. **So, point 1 is correct.**
- **Marine Cloud Brightening** (or Cloud Reflectivity Enhancement) aims to increase the whiteness of clouds in order to reflect more sunlight back into space. Proposals include **spraying a fine mist of seawater** into low-lying marine clouds to create more cloud condensation nuclei, which would allow more vapour to condense and brighten the clouds. **So, point 2 is correct.**
- Reservoirs serve as crucial water regulation mechanisms in arid areas, playing a vital role in meeting the water demand. High evaporation in plain reservoirs in arid regions exacerbates water shortages. **Shade balls, also known as evaporation balls** or bird balls, are plastic spheres that float on water to reduce evaporation. The floating plastic balls cover the water's surface to block sunlight, **reducing evaporation by 85–90 percent** during a drought. By **reducing the absorption of sunlight by the water**, this technique indirectly helps reflect more sunlight back, though its primary aim is to reduce water evaporation. **So, point 3 is correct.**
- **Painting roofs with reflective materials**, known as cool roofing, **enhances the albedo** of buildings by reflecting more sunlight, which helps lower urban temperatures. Applying heat reflective coating for the roof helps in reflecting the sun's rays away, maintaining a cooler indoor environment, and reducing the energy needed for air conditioning. **So, point 4 is correct.**
- **Sprinkling diamond dust, which** is part of solar radiation management, aims to increase the reflectivity of the Earth's atmosphere by dispersing small particles that reflect sunlight back into space. It could offset almost all the warming caused by humans since the industrial revolution and buy us some time with climate change. **So, point 5 is correct.**

Therefore, option (d) is the correct answer.

Q52.
Answer: a
Explanation:

- According to the World Meteorological Organization, the **a**verage annual mean temperature of the Earth during the year was more than **1.5 degrees Celsius higher than pre-industrial levels**

(1850-1900 period). However, according to the India Meteorological Department (IMD), the temperature over **India in 2024 was about 1.2 degrees** Celsius higher than the 1901-1910 average. [*IMD does not have temperature data over India for the 1850-1900 period to facilitate a straight comparison as its dataset begins from 1901*]. **So, statement 1 is correct.**

- **India is located in the tropical area,** quite close to the equator. The increase in global temperature has been more pronounced in the higher altitudes**, near the polar regions, than near the equator**. This is attributable to a complex set of atmospheric phenomena, including **heat transfers from the tropics to the poles** through prevailing air circulation systems. **So, statement 2 is correct.**
- Also**, greater concentration of aerosols, or suspended particles, in the atmosphere over the Indian region also impacts temperature rise**. Aerosols generally have a cooling effect because they scatter the solar radiation back into space. Aerosols also affect cloud formation, which, in turn, affects how much sunlight is absorbed or reflected. **So, statement 3 is correct.**

So, both Statement–II and Statement–III are correct and both of them explain Statement–I. Therefore, option (a) is the correct answer.

Relevance: India has recently launched Mission Mausam to expand its weather observation network and strengthen computing and analysis capabilities.

Q53.

Answer: d

Explanation:

- **Diego Garcia** is a coral atoll, the **largest** and southernmost member **of the Chagos Archipelago** in the central Indian Ocean. It is part of the **British Indian Ocean Territory (BIOT)** and accounts for more than half the land area**.**
- Discovered by the Portuguese in the early 16th century, Diego Garcia was a dependency of Mauritius for most of its history. In 1965 it was separated from Mauritius as part of the newly created BIOT. **In 1966, the UK leased Diego Garcia to the US to create an air and naval base.** To construct the defense installation, the island's inhabitants were forcibly removed.
- **In 2019, the International Court of Justice ruled in an advisory opinion that Britain's decolonization of the island was not lawful because of continued Chagossian claims**. A non-binding 2019 UN General Assembly vote demanded that Britain end its "colonial administration" of the Chagos Archipelago and return it to Mauritius.

Therefore, option (d) is the correct answer.

Relevance: Recently, 15 fishermen from Tamil Nadu were detained near Diego Garcia Island for allegedly crossing the maritime boundary.

Q54.

Answer: d

Explanation:

- ***Trichophyton (T.) indotineae* is a newly identified fungal species that has been found in a near-epidemic form on the Indian subcontinent and caused skin infection**. *T. indotineae* has become a problematic dermatophyte due to its resistance to most anti-fungal medicines, including terbinafine.
- It has mostly been reported to spread person-to-person through skin-to-skin contact or shared objects, causing relapsing infections in South Asia.

Therefore, option (d) is the correct answer.

Relevance: Dermatologists from India and 13 other countries, including Germany, have objected to the region-specific nomenclature given to Trichophyton indotineae because it leads to considerable misinformation, stigma, and racial prejudice.

Q55.
Answer: c
Explanation:

- **Gaemi was** formed as a tropical depression in the **Philippine Sea o**n July 19th, 2024, and then strengthened into a tropical storm. Gaemi made its landfall in Taiwan and China, and flooding from Gaemi caused massive **displacement and deadly mudslides in China. So, pair 1 is not correctly matched**.
- **Rafael was a hurricane in Cuba that landed in the western Cuban province of Artemisa in November 2024**. Approaching Cuba, it traversed warm waters and encountered light to moderate vertical wind shear—environmental conditions that helped the storm to strengthen into a major hurricane. **So, pair 2 is correctly matched**.
- **Hurricane Helene** was a devastating tropical cyclone that caused widespread catastrophic damage across the **Southeastern United States** in late September 2024. It was the strongest hurricane on record to strike the Big Bend region of Florida. **So, pair 3 is not correctly matched.**
- **Dikeledi** was a long-lived tropical cyclone that traversed the southern Indian Ocean in December 2024 and January 2025. It formed south of Java, **Indonesia,** as a tropical low on 30 December 2024, traversing the southern Indian Ocean before entering the South-West Indian Ocean basin. Dikeledi made **landfall near Antsiranana, Madagascar. So, pair 4 is correctly matched.**

Therefore, option (c) is the correct answer.
Relevance: Recently, India extended humanitarian aid to Cuba after Hurricane Rafael.

Q56.
Answer: a
Explanation:

- In the North American continent, the **tropic of cancer** passes through Hawaii, Mexico, the **Gulf of Mexico,** the Gulf of California, the Strait of Florida, and the Bahamas**. So, statement 1 is correct**.
- The Gulf of Mexico is a semi-enclosed sea connected to the **Caribbean Sea to the south through the Yucatan Channel** and the **Atlantic Ocean** in the east through the **Strait of Florida**. The circulation in the Gulf of Mexico is dominated by the Caribbean Current/Loop Current/Florida Current system and the eddies of the Loop Current. These mesoscale circulations and their variability were key in distributing oil and oil dispersants in the GoM. **So, statement 2 is not correct.**
- **Gulf of Mexico** is bounded on the northeast, north, and northwest by the Gulf Coast of the United States; on the southwest and south by the Mexican states of Tamaulipas, Veracruz, Tabasco, Campeche, Yucatán, and Quintana Roo; and on the **southeast by Cuba**. The coastal areas along the Southern U.S. states of Texas, Louisiana, Mississippi, Alabama, and Florida, which border the gulf on the north, are occasionally referred to as the "Third Coast" of the United States. **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.
Relevance: US President Donald Trump said that he would change the name of the Gulf of Mexico to "Gulf of America."

Q57.

Answer: c
Explanation:

- **Human cells contain two sets of the genome:** one derived from the mother's egg and the other from the father's sperm. Eggs and sperm receive only one chromosome of each pair. The latter has two sets again when they fuse and form the zygote. The zygote then multiplies to form a baby.
- When a gene is 'expressed'**, a cell transcribes the underlying base pair sequence to a messenger RNA (mRNA) molecule.** These mRNA, in turn, code for the amino-acid sequences in a polypeptide, which ultimately forms a protein. **So, statement 1 is correct.**
- **Any change in the DNA sequence of a cell is called a mutation**. Mistakes during cell division may cause mutations, or they may be caused by exposure to DNA-damaging agents in the environment. Mutations can be harmful, beneficial, or have no effect. If they occur in cells that make eggs or sperm, they can be inherited; if mutations occur in other types of cells, they are not inherited. Specific mutations may lead to cancer or other diseases. A mutation is sometimes called a variant. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.
Relevance: In December 2024, two research groups, one from the University of Nevada and another from the Max Planck Institute for Evolutionary Biology, addressed how new genes are created.

Q58.
Answer: b
Explanation:

- The **Hoollongapar Gibbon Sanctuary,** formerly known as the Gibbon Wildlife Sanctuary or Hollongapar Reserve Forest, is an isolated protected evergreen forest area located in the **Jorhat district of Assam, India**. The Hoollongapar Gibbon Sanctuary contains India's only gibbons – the hoolock gibbons, and Northeastern India's only nocturnal primate – the Bengal slow loris.
- **The National Board for Wildlife** has recently approved a proposal to carry out **oil and gas exploration in the eco-sensitive zone** of the Hollongapar Gibbon Wildlife Sanctuary. The Hollongapar Gibbon Wildlife Sanctuary spans 20.98 sq km, while its ESZ covers 264.92 sq km. The large ESZ ensures connectivity between the sanctuary, the Dissoi Valley Reserve Forest, and forested areas in Nagaland. This connectivity is crucial for the seven species of primates in the area. Cairn Oil & Gas, a subsidiary of the Vedanta Group, led the proposed oil exploration project.

Therefore, option (b) is the correct answer.
Relevance: Oil and gas exploration will likely threaten Hoolock Gibbon's habitat in Assam.

Q59.
Answer: c
Explanation:

- Tax devolution refers to the **distribution of tax revenues** between the Central Government and the State Governments. It is a constitutional mechanism to allocate the proceeds of certain taxes among the Union and the States fairly and equitably. The Finance Commission recommends the division of net proceeds of taxes between the union and the states.

Therefore, option (c) is the correct answer.
Relevance: The Union Government has disbursed ₹1,73,030 crore to state governments as part of tax devolution. This marks a rise from the ₹89,086 crore distributed in December 2024.

Q60
Answer: c
Explanation:

- The winter of 2024 was an exceptionally dry season for southern California. **Santa Ana winds,** which are common in California at this time of the year, were unusually strong in 2024. Thus, once a fire started in the prevailing dry conditions, the strong **Satna Ana winds made the flames get bigger and spread faster. So, statement 1 is correct.**
- **Santa Ana** winds are a type of **katabatic wind, which are winds that flow downhill.** Between October and January, **high pressure builds up in the deserts of the Great Basin.** The high-pressure system of the Great Basin circulates clockwise and brings wind westwards, towards the coast — different from the usual system of wind blowing inland from the sea. This wind blows down over the Sierra Nevada and Santa Ana mountains, **losing humidity and heating up**. As it is squeezed through mountain passes, it gains much speed. Thus, when it blows over southern California, the wind is fast, dry, and warmer. **It is this wind system that has caused the California fire. So, statement 2 is not correct.**

So, Statement–I is correct, but Statement–II is incorrect.
Therefore, option (c) is the correct answer.
Relevance: The Southern California wildfire has caused the death of 29 people and is considered the second most devastating fire in California history, sparking major debate on the linkage between climate change and natural disasters.

Q61.
Answer: b
Explanation:

- The **Makran Coast** is a 1,000-kilometer stretch along the Gulf of Oman, extending from **Raʾs al-Kūh in Iran to Lasbela District in Pakistan.** It is characterized by rugged terrain, dramatic cliffs, and sandy beaches, with significant ecological and historical importance.

- Recently, Iran has announced plans to relocate its capital from **Tehran to the Makran coastal region**. The decision, primarily driven by economic and ecological concerns, aims to address Tehran's overpopulation, severe pollution, and environmental challenges.
- The new capital would leverage **Makran's strategic location near the Gulf of Oman**. However, this relocation is still in its exploratory phase and has not yet become an immediate priority for the government.

Therefore, option (b) is the correct answer.

Q62.
Answer: c
Explanation:

- The Kisan Pehchaan Patra (Farmer ID) is an initiative by the Government of India aimed at **digitizing farmers' identities and linking them to state land records.** It plays a crucial role in streamlining the distribution of agricultural benefits and ensuring that only genuine farmers receive financial aid under schemes like PM-KISAN. The integration of this system into the Digital Agriculture Mission is a significant step toward modernizing India's agrarian economy.
- Kisan Pehchaan Patra is a unique digital identity assigned to farmers across India, ensuring authenticity and transparency in agricultural welfare programs. It is linked **dynamically to Aadhaar and state land records,** reducing fraudulent claims and allowing direct targeting of government schemes. By integrating the Farmer ID with land ownership records, the government can **verify farmer eligibility for subsidies, loans, and financial aid programs.** This digital infrastructure also enhances accessibility to crop insurance, credit facilities, and advisory services through a centralized database. **So, statement 1 is correct.**
- The Farmer ID contains extensive details about individual farmers, including demographic information, land ownership status, types of crops cultivated, and livestock holdings. By **incorporating livestock data, the ID plays a critical role in developing targeted animal husbandry programs**, ensuring efficient management of resources for dairy farming and cattle welfare schemes. The inclusion of crop sown data also allows policymakers to analyze agricultural trends, plan procurement strategies, and offer customized support to farmers based on regional crop patterns.
- The Government of India has **made it mandatory for all new applicants under the Pradhan Mantri Kisan Samman Nidhi (PM-KISAN) scheme** to register with the Farmer Registry and provide a Farmer ID starting January 1, 2025, in 10 states. The PM-KISAN scheme provides ₹6,000 per annum to small and marginal farmers through Direct Benefit Transfer (DBT). Making the Farmer ID compulsory helps in **eliminating duplicate beneficiaries**, preventing fraudulent claims, and simplifying the disbursement process. The mandatory requirement ensures that only land-owning farmers receive the benefits, thereby reducing leakages and enhancing financial efficiency. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.

Q63.
Answer: d
Explanation:

- TraceNet 2.0 is an upgraded **online organic traceability system designed to enhance regulatory oversight and provide seamless operations**.
- It aims to ensure end-to-end traceability of organic products, from their production to export. It enhances transparency and allows stakeholders to track organic products through every stage of the supply chain, ensuring compliance with organic certification standards and promoting the integrity of organic products in the market.

- This service is part of the **National Programme for Organic Production (NPOP)**, supporting India's organic certification system.

Therefore, option (d) is the correct answer.

Q64.
Answer: c
Explanation:

- India has undertaken several defense projects to enhance its naval capabilities. Among these, Project 17A, Project 15B, and Project 75 are significant programs aimed at building advanced vessels for the Indian Navy.
- **Project 17A** focuses on the development of **stealth frigates for the Indian Navy.** These frigates are equipped with advanced sensors and weapons, offering improved stealth and versatility for naval operations. The **Nilgiri-class frigates, p**art of this project, are designed to enhance India's maritime defense capabilities. **So, pair 1 is not correctly matched.**
- **Project 15B** is dedicated to building destroyers for the Navy. The Visakhapatnam-class destroyers are part of this project and are equipped with **cutting-edge weaponry, stealth features, and advanced sensors** for high-intensity naval warfare. These destroyers are designed to strengthen India's defense in the Indian Ocean region. **So, pair 2 is not correctly matched.**
- **Project 75** involves the construction of conventional diesel-electric submarines, specifically the **Scorpene-class submarines**. These submarines are among the most advanced in the world, **featuring stealth, modern sonar systems, and the ability to launch cruise missiles.** This project significantly bolsters India's undersea capabilities. **So, pair 3 is correctly matched.**

Therefore, option (c) is the correct answer.

Q65.
Answer: c
Explanation:

- Carnivorous plants developed in regions with **nutrient-deprived soils, primarily lacking nitrogen**. To survive in these environments, they evolved the ability to capture and digest small animals, allowing them to obtain nutrients that are otherwise scarce. **So, statement 1 is correct.**
- Various carnivorous plants, like the Venus flytrap and pitcher plants, capture and digest small insects. This adaptation is essential for **obtaining nutrients such as nitrogen and phosphorus**, which are limited in the soil.
- While some species of carnivorous plants do produce digestive enzymes, others rely on microbial activity or **mechanical digestion to break down their prey.** The Venus flytrap, for example, does not rely on enzymes alone but also mechanical crushing to digest prey. **So, statement 2 is not correct.**
- Many unrelated carnivorous plant species developed similar prey capture mechanisms **due to convergent evolution**. Examples include the **sticky traps in sundews and snap traps in Venus flytraps**, which serve the same purpose of capturing insects but evolved independently in different plant families. **So , statement 3 is correct.**

Therefore, option (c) is the correct answer.

Q66.
Answer: b
Explanation:

- The **Modhweth Festival** is the most significant annual celebration for the **Toda tribe**, residing in the Nilgiris Hills of **Tamil Nadu.** During the festival, the Toda community gathers to offer prayers to the deity Thenkish Amman, seeking blessings for good health, rainfall, and bountiful harvests. **So, row 1 is not correctly matched.**
- The **Hemis Festival in Ladakh i**s one of the most important **Buddhist festivals.** It commemorates the birth anniversary of Guru Padmasambhava, who is believed to have brought Buddhism to Tibet. During the festival, **mask dances and rituals are performed, reflecting the cultural richness of the region**, called "**Chham Dances,**" which symbolize the triumph of good over evil**. So, row 2 is correctly matched.**
- The **Karchi Festival**, also known as Kharchi Puja, is a significant annual celebration in Tripura, dedicated to the worship of the fourteen **deities of the Tripuri tribe.** During the festival, the people of Tripura also **worship the earth** along with their 14 deities. The name 'Kharchi' is derived from the word 'Khya,' which means Earth, highlighting the festival's focus on worshipping the earth and the fourteen gods who form the dynasty deity of the Tripuri people. The **Puja is traditionally performed to wash away sins and cleanse the earth** following its post-menstrual phase. This ancient ritual, steeped in tribal origins, is performed over seven consecutive days in July, starting on the eighth day of the new moon.**So, row 3 is correctly matched.**

Therefore, option (d) is the correct answer.

Q67.
Answer: a
Explanation:

- **KM3NeT Project** is a research infrastructure housing the **next generation neutrino telescopes** with a volume of at least one cubic kilometre. It uses Cherenkov radiation (light that neutrinos produce when they interact with a water or ice molecule) to study neutrinos. As a **European research infrastructure**, it is located in the Mediterranean Sea and involves collaboration among multiple countries.
- The **KM3NeT project** focuses on detecting neutrinos from **astrophysical sources such as supernovae, gamma-ray bursters, and colliding stars**. These high-energy neutrinos are detected using a telescope system, making the project an essential tool for astrophysical research. **So, statement 1 is correct.**
- While both the **KM3NeT and IceCube observatories detect high-energy neutrinos**, **IceCube is not part of the KM3NeT project.** IceCube is located under the Antarctic ice, while KM3NeT operates under the **Mediterranean Sea.** Both projects are separate entities, each using different methodologies to detect neutrinos, but IceCube is not directly involved in KM3NeT. **So, statement 2 is not correct.**

Therefore, option (a) is the correct answer.

Q68.
Answer: a
Explanation:

- **Fire-and-Forget** missiles are designed to require **no further guidance once launched.** These missiles rely on onboard sensors such as infrared, radar, or laser guidance to independently track and strike their targets. This allows the **operator** to fire the missile and immediately take cover or engage other threats **without maintaining visual contact**. **So, statement 1 is correct.**
- In conventional guided missiles, the aircraft that had fired the missile has to stay in the vicinity of the target till the missile strikes. But a **'fire and forget' principle missile has a smart computer** which **decides its own path once the target** is set in it so the pilot doesn't have to wait until the final result. **So, statement 2 is correct.**

So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I. Therefore, option (a) is the correct answer.

Q69.
Answer: b
Explanation:

- **Micro-DNA** refers to small, **circular extrachromosomal DNA fragments** that can form due to **errors in DNA replication or repair.** These fragments are often found in cancerous and aging cells, indicating their link to **genetic instability**. Scientists study micro-DNA to understand mutation hotspots and disease progression, particularly in cancer and neurological disorders. **So, statement 1 is correct.**
- **Messenger RNA (mRNA)** carries genetic instructions from DNA to ribosomes for **protein synthesis**. **Micro-DNA,** however, **does not participate in this process**. Instead, micro-DNA may modulate gene activity indirectly by influencing chromatin structure or interacting with cellular pathways. Its role is crucial in genome maintenance, helping repair errors that occur during cell division. This makes **micro-DNA distinct from both mRNA and non-coding RNAs (like microRNAs)**, which regulate gene expression. **So, statement 2 is correct.**
- Micro-DNA is not the same as microRNA (miRNA). While both are small and involved in genetic regulation, their structures and functions differ. **MicroRNA** is a short RNA molecule that **binds to messenger RNA (mRNA) to silence genes or regulate protein synthesis. Micro-DNA** is a DNA fragment, not an RNA molecule, and **does not directly participate in gene silencing**. While both can influence gene activity, they operate through entirely different biological mechanisms. **So, statement 3 is not correct.**
- Micro-DNA is **detected in abnormal levels in certain cancers, making it a potential biomarker** for early-stage tumor detection and prognosis. Some cancers use micro-DNA to evade immune detection or promote drug resistance. **So, statement 4 is correct.**

Therefore, option (b) is the correct answer.

Q70.
Answer: b
Explanation:

- Project **VISTAAR** (Virtually Integrated System to Access Agricultural Resources) is an initiative launched through a collaboration between **IIT Madras and the Ministry of Agriculture and Farmers' Welfare**.
- This digital platform is designed to **enhance the agricultural extension system** by **integrating information on over 12,000 start-ups** in agriculture and allied sectors. By utilizing digital technology, Project VISTAAR aims to create an accessible database that connects farmers with **technological solutions, advisory services, and start-up innovations**. The initiative seeks to **bridge the gap between farmers and agricultural advancements**, ensuring that modern farming techniques, tools, and government schemes reach the grassroots level efficiently.
- The project holds significant implications for India's agriculture sector. By digitalizing the agricultural extension system, farmers can access real-time advisory services on **crop production, marketing, value addition, and supply chain management**.
- Moreover, it provides **detailed information on government schemes**, ensuring that farmers benefit from subsidies and rural development initiatives. This initiative will increase productivity, reduce information asymmetry, and improve decision-making for farmers.

Therefore, option (b) is the correct answer.

Q71.

Answer: a
Explanation:

- The Indian government has launched the **"One Nation, One Subscription" (ONOS) scheme**, aiming to provide seamless **access to a vast array of digital knowledge resources,** including research papers, journals, and academic content, under a single subscription platform.
- This initiative seeks to **benefit students, researchers, and institutions nationwide by eliminating multiple subscription barriers** and ensuring equitable access to knowledge.
- **Key features of the ONOS scheme include:**
    - **Access to International Journals:** Over 13,400 international journals covering various disciplines, including science, technology, engineering, medicine, mathematics, management, social sciences, and humanities, will be made available to researchers.
    - **Beneficiaries:** The scheme will initially benefit over 1.8 crore students from government-funded higher education institutions, including universities and IITs.
    - **Expansion Plans:** Future phases of ONOS will focus on expanding access to private higher educational institutions and ultimately to all individuals in the country who require such access.
    - **Budget Allocation:** The ONOS scheme has a budget of ₹6,000 crore for a period of three years, with an additional ₹150 crore per year allocated for supporting beneficiary authors in publishing research papers in selected Open Access journals.

Therefore, option (a) is the correct answer.
Relevance: Government has launched ONOS to provide seamless access to digital knowledge resources rephrase

Q72.
Answer: d
Explanation:

- **Diammonium phosphate (DAP)** is the world's most widely used phosphorus fertilizer. It is formulated in a **controlled reaction of phosphoric acid with ammonia,** where the hot slurry is then cooled, granulated and sieved. **Because of its high nutrient content and excellent physical properties,** it is a popular choice in farming and other industries. Following are some of the applications of Diammonium Phosphate.
    - **Fertilizer Industry:** DAP fertilizer is an excellent source of **phosphorus (P) and nitrogen (N) for plant nutrition**. It is highly soluble and thus dissolves quickly in soil to release plant-available phosphate and ammonium. **So, point 1 is correct.**
    - **Wine Production: Yeast needs nitrogen to ferment wine,** converting grape sugars into alcohol. Di-Ammonium Phosphate (DAP) provides this essential nitrogen, stimulating fermentation and supporting yeast growth. DAP also ensures consistent flavor, eliminating sulfur odors caused by stressed or stuck yeast. **So, point 2 is correct.**
    - **Dairy Production:** DAP is a rich source of phosphorus, a key element needed for **bacterial growth in cheese cultures.** Similar to its use in winemaking, DAP can be added to milk to sustain the fermentation process by supporting the activity of the desired bacteria. **So, point 3 is correct.**
    - **Fire retardant:** A **mixture of DAP and other ingredients** can be spread in advance of a fire to prevent a forest from burning. It then becomes a nutrient source after the danger of fire has passed. **So, point 4 is correct.**
    - **Treatment of wastewater:** DAP is needed to provide the phosphorus concentration to maintain a **healthy microbial population** to break down the organic matter in the waste water. **So, point 5 is correct.**

Therefore, option (d) is the correct answer.

Relevance: In January 2025, the Union Cabinet extended the one-time special subsidy on Di-Ammonium Phosphate (DAP) fertilizer to ensure farmers can get it at affordable prices.

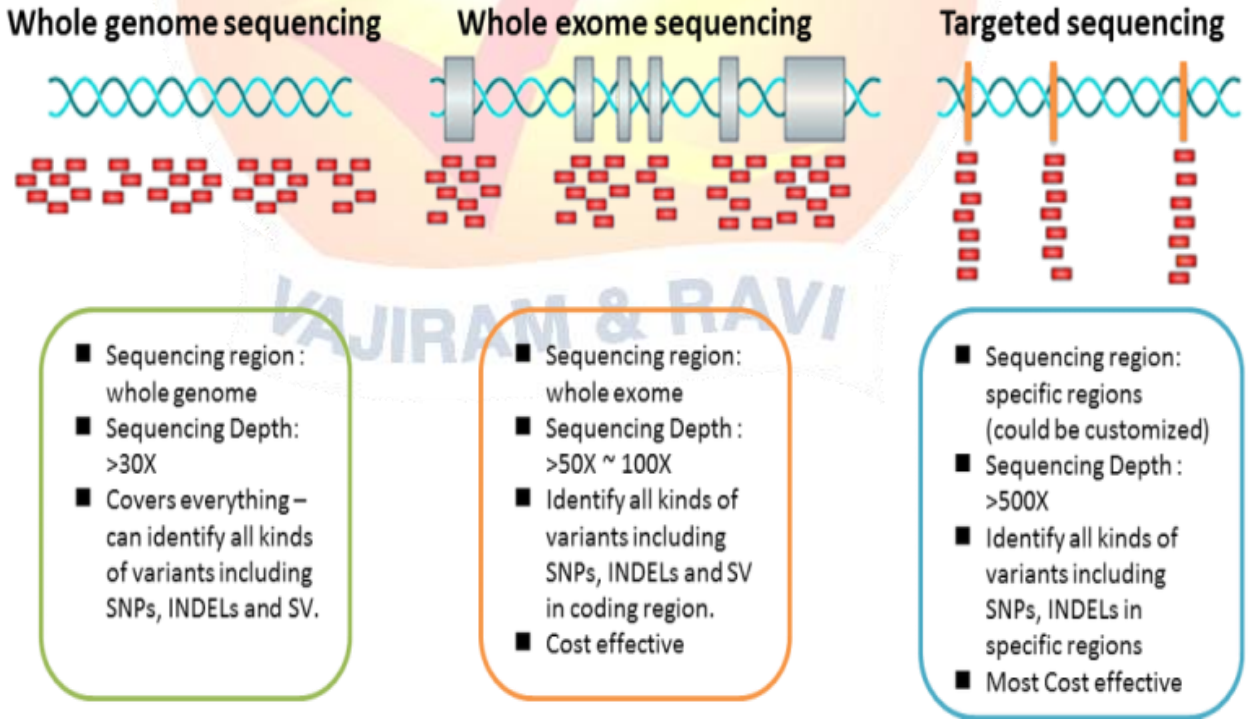
Q73.
Answer: b
Explanation:

- **Pallas's cats, or '*Otocolobus manul*'** are **small wild felids (mammals)** that are the main inhabitants of the steppes of Central Asia. They are also found in western Iran, Mongolia, China, Russia (on the border of Mongolia and China), Kazakhstan, and Kyrgyzstan**.** In these regions, these cats inhabit **mountainous plains and semi-desert foothills**.
- **Pallas's cat is named after Peter Simon Pallas,** a naturalist who first described the species in 1776, based on specimens found near Lake Baikal in Russia. The Pallas's cat is a small, soft-furred animal resembling a house cat. According to the IUCN red list, its status is **near threatened.**
- The species witness declining population due to habitat loss, depletion of prey, and hunting. The **recent sighting of Pallas's cat in Himachal Pradesh** is a major breakthrough, expanding our knowledge of its range and showcasing the Indian Himalayas' incredible biodiversity.

Therefore, option (b) is the correct answer.
Relevance: A groundbreaking photograph of Pallas's cat in Himachal Pradesh, India, captured during a snow leopard survey.

Q74.
Answer: c
Explanation:

- **Genome Sequencing** is the process of determining the complete DNA sequence of an organism's genome, revealing the exact order of its genetic building blocks (nucleotides).



- Whole Genome Sequencing (WGS) and Whole Exome Sequencing (WES) are two distinct approaches in genome sequencing with following differences.

Whole Genome Sequencing (WGS)	Whole Exome Sequencing (WES)
Sequences the **entire genome, coding and non-coding regions of a DNA**, providing a comprehensive view of an individual's genetic information. **So, statement 1 is correct.**	Focuses on protein-coding regions (1-2% of the genome), targeting specific disease-associated variants.
Comprehensive, detects structural variations and regulatory elements.	Targeted, identifies disease-associated variants in protein-coding regions.
WGS detects a wider range of genetic variations and is even more sensitive than WES	WES has comparatively less sensitivity for detecting genetic variations. **So, Statement 2 is correct.**
WGS can identify both rare and common genetic variations, providing a more complete picture of an individual's genetic makeup.	WES can identify rare variants only.

- Both Whole Genome Sequencing and Whole Exome Sequencing enables researchers to explore the genetic basis of health and disease, shedding light on ancestry, evolution, and rare/complex disease causes.
- Next-Generation Sequencing (NGS) technique enables rapid and accurate analysis of both approaches. Both WES and WGS enable discovery of novel genetic associations and personalized medicine (pharmacogenomics) for tailored treatment plans.

Therefore, option (c) is the correct answer.

Q75.
Answer: d
Explanation:

- **The 2024 Annual Groundwater Quality Report** shows promising results, with 81% of groundwater samples deemed suitable for irrigation. Notably, all groundwater samples from North-Eastern states received an "excellent" rating for irrigation, highlighting the region's favorable agricultural conditions.
- This **achievement is attributed to the joint efforts of state and central governments.**
- India has launched **numerous initiatives to promote water conservation and management. Key programs include:**
  - **Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA):** Focuses on water conservation and harvesting structures, enhancing rural water security. **So, Point 1 is correct**.
  - **15th Finance Commission Grants:** Provides financial assistance to states for rainwater harvesting and water conservation.
  - **Mission Amrit Sarovar:** Creates or rejuvenates water bodies for harvesting and conservation. **So, Point 2 is correct**.
  - **Atal Mission for Rejuvenation and Urban Transformation (AMRUT)2.0:** Supports rainwater harvesting and promotes groundwater recharge. **So, Point 3 is correct.**
  - **Atal Bhujal Yojana:** Targets water-stressed areas, focusing on groundwater management.

- **Pradhan Mantri Krishi Sinchayee Yojana:** Aims to expand irrigation coverage and improve water use efficiency.
- **National Aquifer Mapping:** Supports groundwater recharge and conservation plans.
- **Jal Shakti Abhiyan:** Launched in 2019, now in its 5th phase, emphasizing rainwater harvesting and water conservation. **So, Point 4 is correct.**
- **National Water Policy:** Advocates rainwater harvesting and conservation.
- **Watershed Development Component:** Focuses on rainfed and degraded lands, incorporating conservation activities.
- **National Water Awards:** Recognizes exceptional contributions to water conservation and management.

Therefore, option (d) is the correct answer.

Relevance: The 2024 Annual Groundwater Quality Report was released emphasizing the importance of not only preserving groundwater but also ensuring its quality for effective, long-term water management.

Q76.

Answer: a

Explanation:

- The Red Sea is an arm of the Indian Ocean, situated between northeastern Africa and the Arabian Peninsula. The sea branches into the Gulf of Suez (west) and Gulf of Aqaba (east) at its northern end.

- **The Red Sea is a vital waterway, connecting to the Mediterranean Sea via the Suez Canal.** The Red Sea is one of the warmest bodies of water due to its location between the Sahara and Arabian Deserts. An estimated 12-16% of global trade passes through the Red Sea and Suez Canal.
- **There are around 15 major ports, with Saudi Arabia's Jeddah Islamic Port being the busiest.**

- Some of the important ports include,
    - **The Port of Eilat** is **Israel's** sole Red Sea port, handling cargo, containers, and passengers, while also serving as a strategic naval base. Eilat is a popular tourist destination, known for its coral reefs and marine life. **So, pair 1 is not correctly matched.**
    - **The Port of Aqaba** is **Jordan**'s sole maritime gateway, serving as a key logistics hub and driving the country's economy. The port's Aqaba Container Terminal is one of the Red Sea's busiest. **So, pair 2 is not correctly matched.**
    - **Jeddah Islamic Port** is **Saudi Arabia's** primary port and the Arab world's second largest and busiest. **So, pair 3 is correctly matched.**
    - **The Port of Djibouti** is a strategic trading hub and main transshipment point for landlocked **Ethiopia,** handling various cargo types. It's also a key location for international military bases.
    - **The Port of Suez**, **Port of Safaga**, Port of Aden, Port of Mokha are some of the other important ports in the region.

So, only one of the above pairs is correctly matched.
Therefore, option (a) is the correct answer.

Q77.
Answer: c
Explanation:

- Nitrate levels in groundwater have been increasing globally, becoming a growing concern worldwide. **56% of India's districts have excessive nitrates** —Rajasthan, Karnataka, and Tamil Nadu reported the maximum nitrate contamination problem.
- **The permissible limit of nitrate concentration is 45 mg per litre** in the groundwater. Anything more than 45 mg/l is considered excess. **So, statement 1 is not correct.**
- **Excessive nitrate levels in the groundwater is attributed to** natural and human-induced factors associated with agricultural, industrial, and sewage sources.
    - **Man-made factors** include the manufacture and use of nitrate fertilizers, fossil fuel combustion, and releases of both domestic and industrial sewage systems and modification in nitrogen-fixing crops in natural vegetation.
    - **Natural factors** contributing to excess nitrates in groundwater are nitrogen-rich geologic deposits, organic matter decomposition, forest vegetation leaching, lightning strikes and bacterial nitrification.
- Groundwater contaminated with excess nitrates can cause many diseases, including methemoglobinemia, which is also known as **blue baby syndrome,** and stomach cancer in adults.
- Excess nitrates found in groundwater is largely due to subsidised fertilizer use. Central and southern Indian states like Rajasthan, Karnataka, T.N are found with highest nitrate levels.
- **Monsoon rainfall tends to increase nitrate levels** in groundwater by leaching fertilizers from soil, flushing accumulated nitrates into water sources especially affecting agricultural areas with high fertilizer use. **30.77% o**f samples were contaminated in the pre-monsoon period compared to **32.66% after the rainy season**. **So, statement 2 is not correct.**

Therefore, option (c) is the correct answer.
Relevance: As of 2023, a report by the Central Groundwater Board (CGWB) reveals that 440 districts in India have excessive nitrate levels in their groundwater.

Q78.
Answer: b
Explanation:

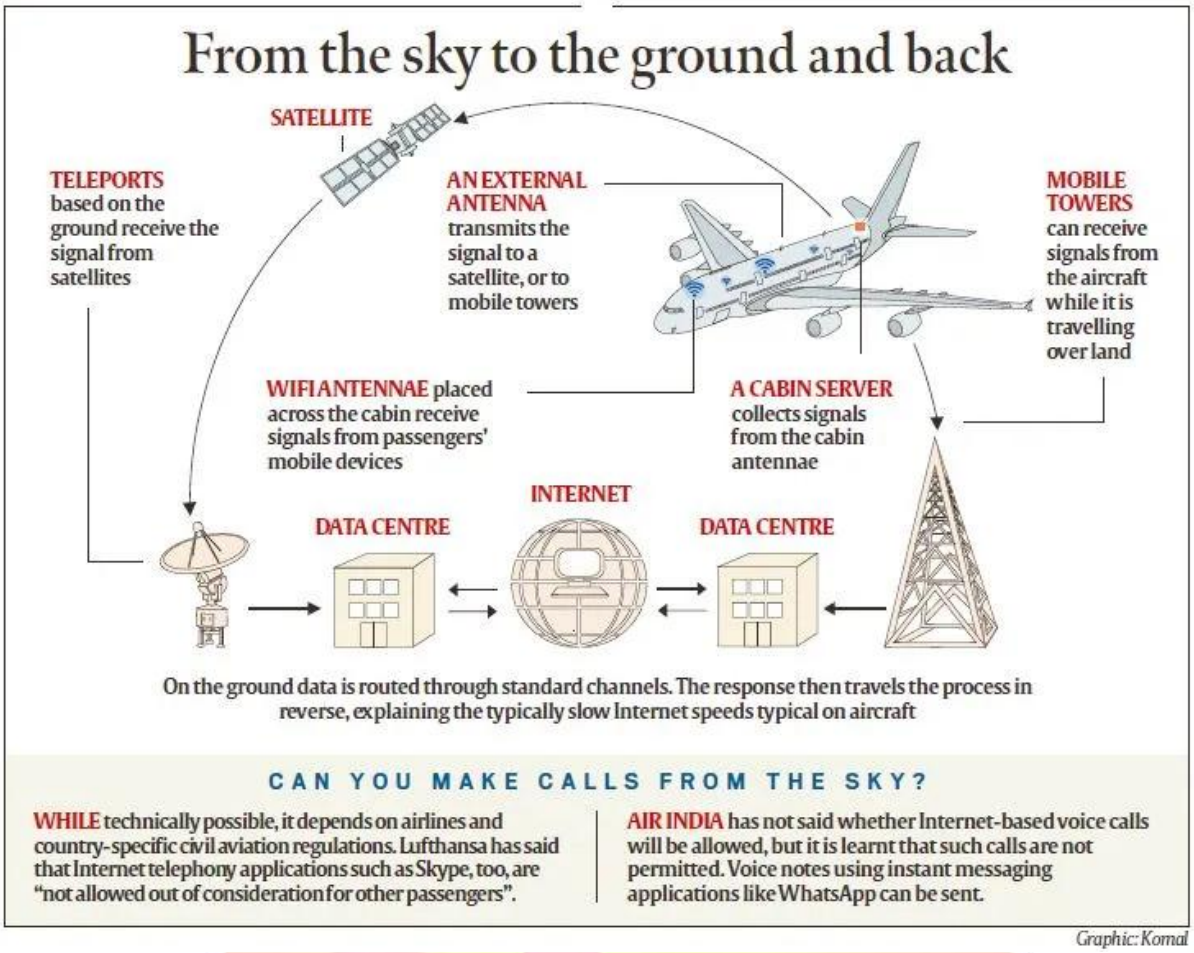
- **Sree Narayana Guru (1856-1928)** was a spiritual leader, philosopher, and social reformer from Kerala, India. He played a pivotal role in transforming the social and cultural landscape of Kerala, particularly among the marginalized Ezhava community.
- Narayana Guru's teachings emphasized the **importance of self-realization, compassion, and social equality.** He advocated for the abolition of the caste system and promoted the **idea of "one caste, one religion, one God for mankind."**
- He encouraged the **Ezhava community to abandon their traditional occupation of toddy tapping**, which was considered impure, and instead adopt more respectable professions. He advocated for **women's education, property rights, and social equality.**
- Narayana Guru was a prolific writer and poet. His notable works include:
    - ***Atmopadesa Satakam*:** A collection of 100 poems that convey spiritual and philosophical teachings.
    - ***Daiva Dasakam*:** A series of poems that describe the nature of God and the universe.
    - ***Anukampa Dasakam*:** A collection of poems that emphasize the importance of compassion and empathy.

Therefore, option (b) is the correct answer.
Relevance: The ruling CPI(M) in Kerala and BJP have locked horns now over Sree Narayana Guru. Under contention are Chief Minister Pinarayi Vijayan's remarks that the towering social reformer and spiritual leader was not a proponent of "Sanatan Dharma".

Q79.
Answer: d
Explanation:

- **In-flight internet connectivity,** also known as in-flight Wi-Fi, is a technology that allows airline passengers to access the internet while flying. In-flight connectivity systems rely on two main technologies: **air-to-ground (ATG) systems**, which utilize ground-based cellular towers, and **satellite-based connectivit**y. To enable these systems, aircraft must be equipped with specialized antennae and equipment.
- **ATG Networks:** This **uses cellular towers on the ground to transmit internet signals**. A small antenna, usually on the plane's belly, receives these signals. The signal is then routed to an onboard router, which distributes the internet connection to passengers' devices. This process is **similar to how devices access Wi-Fi on the ground**. **So, statement 1 is not correct.**
- **Satellite-Based Connectivity:** Satellites in geostationary orbit around the Earth transmit **internet signals from ground stations to aircraft**. A satellite antenna, usually on top of the plane, receives these signals. The signal is then routed to an onboard router, which distributes the internet connection

to passengers' devices. **So, statement 2 is correct.**



Graphic: Komal

So, Statement–I is incorrect, but Statement–II is correct.
Therefore, option (d) is the correct answer.

Q80.
Answer: a
Explanation:

- **The United Nations Framework Convention on Climate Change (UNFCCC)** is an international treaty that aims to stabilize greenhouse gas concentrations in the atmosphere at a level that prevents dangerous anthropogenic interference with the climate system.
- The UNFCCC **categorizes countries into three main groups based on their commitments:**
- **Annex I Parties: Industrialized countries that were OECD members in 1992**, plus countries with economies in transition, including Russia, the Baltic States, and several Central and Eastern European States. **So, pair 1 is not correctly matched.**
    - Under the UNFCCC, the target for Annex I countries is to stabilize their greenhouse gas emissions at 1990 levels, essentially meaning they should aim to not increase their emissions beyond what they were emitting in 1990;
- **Annex II Parties:** OECD members of Annex I, excluding countries with economies in transition. These **countries must provide financial resources to developing countries** to reduce emissions and adapt to climate change. They must also promote the development and **transfer of environmentally friendly technologies**. **So, pair 2 is correctly matched.**
- **Non-Annex I Parties: Mostly developing countries**, including those vulnerable to climate change impacts, such as countries with low-lying coastal areas or prone to desertification and drought. The

UNFCCC emphasizes supporting vulnerable countries through investment, insurance, and technology transfer. These countries are **not assigned specific emission reduction targets** like Annex I countries, but are encouraged to take mitigation actions based on their capabilities. **So, pair 3 is correctly matched.**

So, only one of the above pairs is correctly matched.
Therefore, option (a) is the correct answer.

Q81.
Answer: a
Explanation:

- **Core inflation** is a measure of inflation that **excludes volatile components such as food and fuel**. Since food and fuel prices are **highly sensitive to supply shocks**, seasonal changes, and international market fluctuations, their exclusion results in a **smoother and more stable inflation trend**. **So, statement 1 is correct.**
- **Core inflation is derived from headline inflation** by removing food and fuel components, which tend to be highly volatile. This is done to provide a clearer picture of long-term inflation trends that are primarily **driven by demand-side factors** rather than temporary supply shocks. **So, statement 2 is correct.**
- The exclusion of food and fuel items from core inflation directly contributes to its stability. Since these components are the most volatile, their removal makes core inflation less prone to sudden fluctuations**.**
- **Headline inflation** refers to the **change in value of all goods in the basket**. In a developed economy, food & fuel account for 10-15% of the household consumption basket and in developing economies it forms 30-40% of the basket. Thus headline inflation is more **relevant for developing economies than developed economies**.

So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I.
Therefore, option (a) is the correct answer.

Q82.
Answer: b
Explanation:

- The **Grand Ethiopian Renaissance Dam (GERD)** is a large-scale hydroelectric project on the **Blue Nile River in Ethiopia**. GERD has the largest hydropower generation in Africa, and it is expected to generate up to ~6000 MW of electricity annually. It has been a point of geopolitical tension with downstream countries Sudan and Egypt, which depend on the Nile for water resources. **So, pair 1 is correctly matched.**
- The **Motuo Mega-Dam** is a planned **Chinese** hydropower project on the **Yarlung Tsangpo River (Brahmaputra in India**). China has approved the construction of what will be the **world's largest hydropower dam**, stoking concerns about displacement of communities in Tibet and environmental impacts downstream in India and Bangladesh. The dam, which will be located in the lower reaches of the Yarlung Tsangpo river, could generate **three times more energy than the Three Gorges Dam. So, pair 2 is not correctly matched.**
- The **Shahtoot Dam** is an **Indo-Afghan collaborative project on the Maidan River**, a tributary of the Kabul River. The project aims to provide drinking water for Kabul city and enhance irrigation in Afghanistan. **So, pair 3 is correctly matched.**

So, only two of the above pairs are correctly matched.

Therefore, option (b) is the correct answer.

Q83.
Answer: a
Explanation:

- **Human Metapneumovirus (HMPV)** is a respiratory virus that **typically circulates during late winter and early spring,** similar to other **seasonal respiratory viruses** like influenza and Respiratory syncytial virus (RSV). It can cause cold-like symptoms and severe infections in vulnerable populations, such as young children, the elderly, and immunocompromised individuals. **So, statement 1 is correct.**
- Viruses like HMPV, influenza, and other respiratory pathogens **tend to thrive in cold and dry conditions.** Low humidity helps **viral particles remain airborne for longer, increasing transmission rates.** Additionally, people spend more time indoors during winter, facilitating close contact and the spread of infections. **So, statement 2 is correct.**
- The **dry and cold conditions of winter** create an environment conducive to the spread of respiratory viruses, including **HMPV**. As a result, infections peak during late winter and early spring.

So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I.
Therefore, option (a) is the correct answer.

Q84.
Answer: d
Explanation:

- **Velu Nachiyar (1730-1796)** was a pioneering figure in India's resistance against British colonial rule. As the **queen of Sivagangai**, she led a bold and determined effort to reclaim her kingdom from the British, becoming one of the first Indian rulers to wage war against the British East India Company. Her legacy is marked by her strategic military acumen, alliances, and innovative warfare tactics.
- Velu Nachiyar, born in 1730 in Ramnathpuram, was the **first queen in India to actively resist British rule**. She launched a **military campaign against the British East India Company before the 1857 Revolt**, making her one of the earliest figures in India's anti-colonial struggle. She successfully reclaimed her kingdom of Sivagangai and ruled for a decade. **So, statement 1 is correct.**
- She sought **military support from Hyder Ali** of Mysore to strengthen her campaign against the British. Using innovative military strategies, she orchestrated a **suicide attack on British ammunition depots**, which significantly weakened their position. This attack is considered one of the earliest recorded instances of a human bomb in warfare. **So, statement 2 is correct.**

Therefore, option (d) is the correct answer.

Q85.
Answer: d
Explanation:

- The United Nations Security Council (UNSC) is a key body responsible for maintaining international peace and security. It consists of 15 members, including five permanent members with veto power and 10 non-permanent members elected for two-year terms.
- The five permanent members of the **UN Security Council (P5) – the United States, the United Kingdom, Russia, China, and France – have veto power**, giving them greater influence over decisions compared to the 10 non-permanent members. Thus, voting power is not equal among all members. **So, statement 1 is not correct.**
- Non-permanent members of the UNSC serve for a **term of two years,** not three. They are elected by the General Assembly and rotate every two years. **So, statement 2 is not correct.**

- The Security Council meets as needed **throughout the year to address global security issues,** not just once annually. It is one of the few UN bodies that operates continuously. **So, statement 3 is not correct.**
- A **flag installation ceremony** is held at the UN headquarters at the beginning of a non-permanent member's term, and **their flags are removed at the end of their tenure**. This symbolic event marks their entry and exit from the UNSC.

So, none of the above statements are correct.
Therefore, option (d) is the correct answer.

Q86.
Answer: c
Explanation:

- **Corals** are marine invertebrates belonging to the class Anthozoa of the phylum **Cnidaria**. They are **primarily known for forming coral reefs**, which are vital marine ecosystems that support a vast array of marine life.
- Natural coral spawning is a synchronized event that **typically occurs once a year, when corals release eggs and sperm into the ocean for fertilization.** This event is influenced by environmental factors such as temperature, lunar cycles, and tides. **So, statement 1 is correct.**
- Cryo-born corals are developed using cryopreservation techniques, where coral sperm and eggs are frozen and later used for fertilization. These corals are designed to be more **heat-tolerant, helping reefs adapt to rising ocean temperatures** caused by climate change. Scientists believe that deploying millions of these corals onto reefs will aid in conservation and restoration efforts. The world's first cryopreserved baby corals have been successfully transplanted into their natural habitat on the Great Barrier Reef. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.

Q87.
Answer: d
Explanation:

- **Dhanauri wetland** is located in Greater Noida, Gautam Buddha Nagar district, **Uttar Pradesh**, close to the Okhla bird sanctuary and Surajpur wetland. It is nestled within the **floodplains of the Yamuna basin**, within 15 kms of the river Yamuna.
- The wetland, which is mostly composed of marshes, is a significant site for the vulnerable Sarus Crane (Antigone antigone). Besides the Sarus Crane, some of the bird species reported in the Dhanauri wetland are Common teal, Mallard, Northern-Pintail, Greylag geese, Bar-headed geese, Woolly-necked Stork, Black-necked stork, Painted Stork, and Eurasian marsh harrier among others.
- It is classified as an **Important Bird Area by BirdLife International.**

Therefore, option (d) is the correct answer.
Relevance: The principal bench of the National Green Tribunal (NGT) recently directed the Uttar Pradesh government to place on record in four weeks the status of notifying the Dhanauri water body as a wetland.

Q88.
Answer: c
Explanation:

- Autism spectrum disorder is a neurological and developmental condition marked by disruptions in brain-signaling that cause people to behave, communicate, interact, and learn in atypical ways.
- Autism diagnoses in the United States have increased significantly since 2000. Autism is a spectrum disorder with varying manifestations—some individuals have good conversation skills, while others are

nonverbal or highly sensitive to sensory triggers. Some exhibit repetitive behaviors and may require varying levels of support.

- A growing area of research focuses on epigenetics – the interactions between genetic and environmental factors. One theory suggests that a woman's exposure to **air pollution or harmful contaminants before or during pregnancy** may trigger a genetic mutation leading to autism in her child. **So, statement 1 is correct.**
- Gene variations in some people with autism strengthen the evidence for a genetic component.
- There are **no objective tools for diagnosing autism** such as blood tests or brain scans. Instead, diagnoses are made based on **observations and interviews. So, statement 2 is not correct.**
- There are no treatments or cures for autism, nor can it be reversed. However, experts agree that early diagnosis is crucial. Intervention with supportive measures – ideally before age three – is critical for improving cognitive, social and communication skills.
- Such measures can include speech therapy, occupational therapy, social skills training, sensory integration therapy, visual aids, structured routines, individualized education plans, family therapy, and providing a calm and predictable environment.
- Of children meeting criteria for ASD, the true **male-to-female ratio is closer to 3:1**. There appears to be a diagnostic gender bias, meaning that girls who meet criteria for ASD are at disproportionate risk of not receiving a clinical diagnosis. **So, statement 3 is correct.**

Therefore, option (c) is the correct answer.

Relevance: Growing concern of autism spectrum disorder (ASD) amongst Children.

Q89.

Answer: a

Explanation:

- Mountain passes play a crucial role in connecting different regions across mountainous terrains, allowing for easier travel and trade. Each pass is associated with a specific mountain range.
- The **Banihal Pass** is located in the **Pir Panjal Range**. It is an essential route linking the **Jammu** region to the **Kashmir Valley**. The pass serves as a vital transportation link in the region's mountainous terrain. **So, pair 1 is correctly matched.**
- The **Khunjerab Pass** is located in the **Karakoram Range**, not the **Eastern Himalayas**. It forms a strategic border point between **Pakistan** and **China** at a high altitude. This pass provides a crucial route for trade between the two countries. **So, pair 2 is not correctly matched.**
- The **Chang-La Pass** is situated in the **Ladakh Range**, connecting the region of **Leh** to **Nubra Valley**. It is one of the highest motorable passes in the world, at an altitude of around 17,590 feet. This pass is vital for access to strategic regions, including **Siachen Glacier**. **So, pair 3 is correctly matched.**
- The **Nathu La Pass** lies in the **Eastern Himalayas**, not the **Karakoram Range**. It connects **Sikkim** in India to **Tibet** in China. The pass is historically significant and was a key trade route before its closure and later reopening. **So, pair 4 is not correctly matched.**

Therefore, option (a) is the correct answer.

Q90.

Answer: b

Explanation:

- The **PM Young Achievers Scholarship Award Scheme for Vibrant India (PM-YASASVI)**, launched by the **Ministry of Social Justice and Empowerment**, is designed to provide educational support to students from **Other Backward Classes (OBC)**, **Economically Backward Classes (EBC)**, and **Denotified Tribes (DNT)**.

- By consolidating earlier schemes like the Dr. Ambedkar Post-Matric Scholarship and Pre-Matric Scholarship programs, it ensures a streamlined and impactful approach to support students from **Class 9** to **higher education**. The scheme includes scholarships for **school** and **college students**, along with provisions for **hostel facilities** and financial aid for **top-class education**.
- This initiative addresses financial barriers for marginalized communities, enabling them to pursue quality education and contributing to a more **inclusive** and **equitable** society.

Therefore, option (b) is the correct answer.

Q91.
Answer: c
Explanation:

- **The Central Bureau of Investigation (CBI**), as National Central Bureau for Interpol (NCB-New Delhi) in India, **connects all law enforcement agencies in India** to law enforcement agencies in 195 other countries through **Interpol**. **So, statement 1 is not correct.**
    - **Currently, all central agencies and the police forces of various states and Union Territories coordinate with the CBI**, Interpol liaison officers (ILOs), and unit officers concerned through letters, emails, and faxes, due to which they often face delays in their investigation.
- Bharatpol is a broadcast hub for assistance and real-time action against transnational crimes via international police cooperation, **developed by the CBI**. **So, statement 2 is not correct.**
- The decision to develop this portal was taken amid the rise in transnational crimes, including cyber-crime, financial crimes, online radicalisation, organised crimes, drug trafficking, and human trafficking. In such cases, real-time international assistance is needed for criminal investigations. There are **five key features**:
    - **Unified Platform**: This portal integrates the CBI, as the Interpol (NCB-New Delhi), with all law enforcement authorities in India, down to Superintendents of Police (SPs) and Commissioners of Police (CPs).
    - **Simplified Request Mechanism**: This portal allows front-line police officers to easily and promptly request international assistance from 195 Interpol member countries using standardised templates.
    - **Rapid Information Dissemination**: This portal enables the CBI as the NCB to rapidly share criminal intelligence and inputs from 195 countries with all law enforcement agencies in India.
    - **Increase utilisation of Interpol notices**: This portal will **enable easy drafting of Red Corner Notice requests** and other colour coded notices of Interpol. This will lead to effective tracking of crime, criminals and proceeds of crime globally. **So, statement 3 is correct.**
    - **Capacity Building and Training**: This portal also provides access to relevant documents, templates, and training resources, enhancing the capability of frontline officers to conduct investigations abroad and seek foreign assistance effectively through Interpol.
- There are five key modules — Connect, INTERPOL Notices, References, Broadcast, and Resources.

Therefore, option (c) is the correct answer.

Relevance: Union Home Minister, GOI has recently inaugurated the 'Bharatpol' portal to streamline international cooperation for law investigating agencies.

Q92.
Answer: d
Explanation:

- Tidal tails are thin, elongated regions of stars and interstellar gas extending into space. They are typically **formed when galaxies undergo close encounters or mergers**, causing their mutual gravitational forces to distort and stretch out material from their outer regions.
    - During the interaction, **gas and stars** are often **stripped from the outer regions** of the galaxies to **form two tidal tails**: one trailing and one preceding each galaxy.
    - These tails can persist long after the galaxies have finally merged and are therefore considered a **signature of recent merger activity.**
- Examples of galaxies with tidal tails include the Tadpole Galaxy and the Mice Galaxies.
- Within those galaxies which have tidal tails, approximately 10% of the galaxy's stellar formation takes place in the tail**.**

Therefore, option (d) is the correct answer.
Relevance: Astronomers have discovered a newly forming ultra-diffuse galaxy at the tip of a record-breaking tidal tail extending from NGC 3785, located approximately 430 million light-years away in the Leo constellation.

Q93.
Answer: b
Explanation:

- India has **56 Central Universities (CU**) that are recognized for their quality education and research opportunities. These universities operate under the Department of Higher Education, Ministry of Education, and offer a wide range of courses in science, arts, commerce, humanities, law, and technology, contributing significantly to the nation's higher education system.
- They are **established by the Government of India on the recommendation of Parliament as an act** and come under the purview of the Department of Higher Education in the Ministry of Education. **So, statement 1 is not correct.**
- The Government plays a crucial role in the appointment of the Vice-Chancellor (VCs) and the autonomy of the university is given scant regard. **VCs in central universities are appointed by the President of India**, the ex-officio visitor of all central universities, who generally acts on the advice of the Centre. **So, statement 2 is correct.**
- However, according to the SC judgment in Dr Premachandran Keezhoth (2023), he acts purely as a statutory authority under a particular university Act and not as constitutional head of state. In state universities, appointments are made by the Governor.
- The VC as the head of the university is expected to function as a "bridge" between the administrative and the academic wings. The qualifications of VC for the first time were laid down by UGC in 2010.
    - As per the **2018 UGC regulations**, the VC should be a "distinguished academician, with a minimum of 10 years of experience as professor in a university system or 10 years of experience in an equivalent position in a reputed research and/or academic administrative organisation".

Therefore, option (b) is the correct answer.
Relevance: Several states have objected to provisions in the UGC (Minimum Qualifications for Appointment and Promotion of Teachers and Academic Staff in Universities and Colleges and Measures for the Maintenance of Standards in Higher Education) Regulations, 2025.

Q94.
Answer: a

Explanation:

- An **artesian aquifer** stores water under pressure between impermeable layers of sediment and soil below the Earth's surface. It is also referred to as "confined" water due to the presence of hardy materials above and below it.
- These are the **characteristics of artesian aquifer**:
  - The water is confined in layers of rock or soil, creating pressure that forces it to the surface when tapped.
  - Due to the underground pressure, water can flow freely from the well once it is drilled, often without the need for a pump.
  - Artesian wells typically tap into deeper underground sources compared to regular wells.
- Artesian wells are formed **when water from a higher elevation is trapped between two impermeable layers**, and the **pressure from the surrounding rock forces it upward when a hole or well is drilled**.
- Artesian wells are commonly found in areas with specific geological conditions, and they can provide a steady flow of water for various uses, including irrigation and drinking water. This water is located deeper below the surface and is surrounded by poorly permeable rocks that trap the pressure.

Therefore, option (a) is the correct answer.

Relevance: Geologists at Rajasthan's State Ground Water Department, explained the phenomenon as an "artesian condition," where water flows naturally due to underground pressure.

Q95.

Answer: a

Explanation:

- The **Electricity (Rights of Consumers) Rules, 2020**, reduce the timeline for new electricity connections and simplify rooftop solar installation.
- Consumers in multi-storied flats can now choose their connection type, ensuring separate billing for common areas and backup generators. Distribution companies must install check meters for consumer complaints regarding electricity consumption.
- Consumers can now obtain **separate electricity connections for EV charging stations**, aligning with India's **Net Zero** target by 2070. **So, statement 1 is correct.**
- It facilitates faster installation of **Rooftop Solar Photovoltaic systems**. The distribution companies will bear the **cost of distribution system strengthening for systems up to 5 kW.** The timeline for commissioning rooftop solar systems has been **reduced from 30 to 15 days**. **So, statement 2 is not correct.**
- The timeline for obtaining **new electricity connections** has been reduced to 3 days in metropolitan areas, 7 days in other municipal areas, and 15 days in rural areas, while hilly rural areas will retain a 30-day period.
- In **residential societies**, consumers can choose between individual or single-point connections through a transparent ballot conducted by the distribution company, ensuring parity in tariffs. Metering and billing will be done separately for individual consumption, backup power, and common areas.

Therefore, option (a) is the correct answer.

Relevance: The Government of India has recently approved amendments to the Electricity (Rights of Consumers) Rules, 2020.

Q96.

Answer: d

Explanation:

- The **Pravasi Bharatiya Divas (PBD)** celebrated once in two years on **9th January** is a significant event that honours the contributions of the Indian diaspora to their homeland.

- The Pravasi Bharatiya Divas (PBD) Convention was **first** established in **2003** under the government of then Prime Minister **Late Shri Atal Bihari Vajpayee,** as a platform to **recognize and engage** with the **overseas Indian community**.
- The Pravasi Bharatiya Divas is the **flagship** event of the **Ministry of External Affairs**. It is held in different cities, to showcase the diversity and progress of different regions of India.
- Since **2015**, it has evolved into a **biennale event**, with **theme-based conferences** held in the intervening years.
- This format allows for more focused discussions on specific areas of interest and provides opportunities for networking among the global Indian diaspora.

Therefore, option (d) is the correct answer.

Knowledge Box:
• In November 2009, the **United Nations General Assembly (UNGA)** paid tribute to Nelson Mandela by adopting a resolution to make the international community aware of his humanitarian work. • The resolution declared the birthday of South Africa's first democratically elected president, 18 July, Nelson Mandela International Day. • It was also on that day that Madiba called on the people of the world to honour him by helping their communities.

Relevance: The 18th PBD Convention is scheduled for January 8-10, 2025, in Bhubaneswar, Odisha.

Q97.
Answer: d
Explanation:

- BRICS was established in 2009, BRICS was founded on the premise that international institutions were overly dominated by Western powers and had ceased to serve developing countries.
- The bloc has sought to coordinate its members' economic and diplomatic policies, found new financial institutions, and reduce dependence on the U.S. dollar.
- BRICS countries seek to build an alternative to what they see as the dominance of the Western viewpoint in major multilateral groupings, such as the World Bank, the Group of Seven (G7), and the UN Security Council.
- The BRICS includes 10 countries - **Brazil, China, Egypt, Ethiopia, India, Iran, Russian Federation, South Africa, United Arab Emirates**.
- In 2024, BRICS added five new members - Egypt, Ethiopia, Iran, Saudi Arabia, and the United Arab Emirates.
- In 2025, Indonesia officially joined the BRICS as a full member.

So, the correct chronological order in which these countries became members of BRICS is Brazil - South Africa - Iran - Indonesia.
Therefore, option (d) is the correct answer.
Relevance: Recently, Indonesia has officially joined the BRICS as a full member.

Q98.
Answer: b
Explanation:

- **Victor Ambros and Gary Ruvkun** won the 2024 **Nobel Prize in Physiology or Medicine** for d**iscovering microRNAs** and their role in controlling gene expression.

- This pioneering discovery was made using the **roundworm *Caenorhabditis elegans***. This 1-mm long, slender, and transparent **nematode** has been the star of many path breaking discoveries in biology, four of which have won Nobel Prizes.
- *Caenorhabditis elegans* is a **eukaryotic, multi-organ, transparent nematode** that lives in the interstitial water of soil and survives by feeding on microbes. It is a small animal with a transparent body, ~1 mm adult length, and a short life cycle of ~3 days. It consists of 4 larval stages (L1-L4) and adulthood, with a total longevity of around 2–3 weeks at 20 °C.

Therefore, option (b) is the correct answer.

Relevance: Victor Ambros and Gary Ruvkun won the 2024 Nobel Prize in Physiology or Medicine for discovering microRNAs and their role in controlling gene expression.

Q99.

Answer: a

Explanation:

- India has set a target to achieve 50% cumulative installed capacity from non-fossil fuel-based energy resources by 2030 and has pledged to reduce the emission intensity of its GDP by 45% by 2030, based on 2005 levels.
- The integration of **variable and intermittent renewable energy (like solar and wind)** into the energy mix poses challenges for grid stability and uninterrupted power supply. This is due to the varying nature of renewable energy generation, which depends on factors like time, climate, season, and geographic location.
- **Energy Storage Systems (ESS)** can be used for **storing available energy from renewable energy sources and further can be used** during peak hours of the day. ESS can store excess energy generated from renewable sources and **release it during peak demand hours,** ensuring a stable and reliable power supply. **So, statement 2 is correct.**
- The benefits of ESS includes **bringing down the variability of generation in renewable energy sources**, improving grid stability, enabling energy/ peak shifting, providing ancillary support services, enabling larger renewable energy integration, brings down peak deficit and peak tariffs, reduction of carbon emissions, deferral of transmission and distribution capex, energy arbitrage etc. **So, statement 1 is correct.**

So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I.

Therefore, option (a) is the correct answer.

Relevance: India has set a target to achieve 50% cumulative installed capacity from non-fossil fuel-based energy resources by 2030.

Q100.

Answer: c

Explanation:

- **Human-Wildlife conflict** has emerged as a major threat to wildlife populations. The incidence of such conflicts are increasing as a **consequence of habitat fragmentation** and increased human incursions into wildlife spaces, endangering both human and animal life.
- Recently, villagers of **Naugaon, Assam** adopted a novel method to curb this menace. The region known for its **elephant population** was facing persistent wildlife foraging problems, with elephants trespassing into human settlements in search of food.

- The village civil society started an initiative 'Haathi Bandhu' under which they planted **napier grass**. Elephants are known to relish this grass and as a result the straying of elephants towards the farmer's fields decreased. Thus 'Haathi-Bandhu' initiated an innovative way to **tackle man-elephant conflic**t.

Therefore, option (c) is the correct answer.

Relevance: The Prime Minister in his first 'Mann Ki Baat' Episode of 2025 commended the efforts of villagers of Naugaon for adopting the 'Haathi Bandhu' programme.

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